

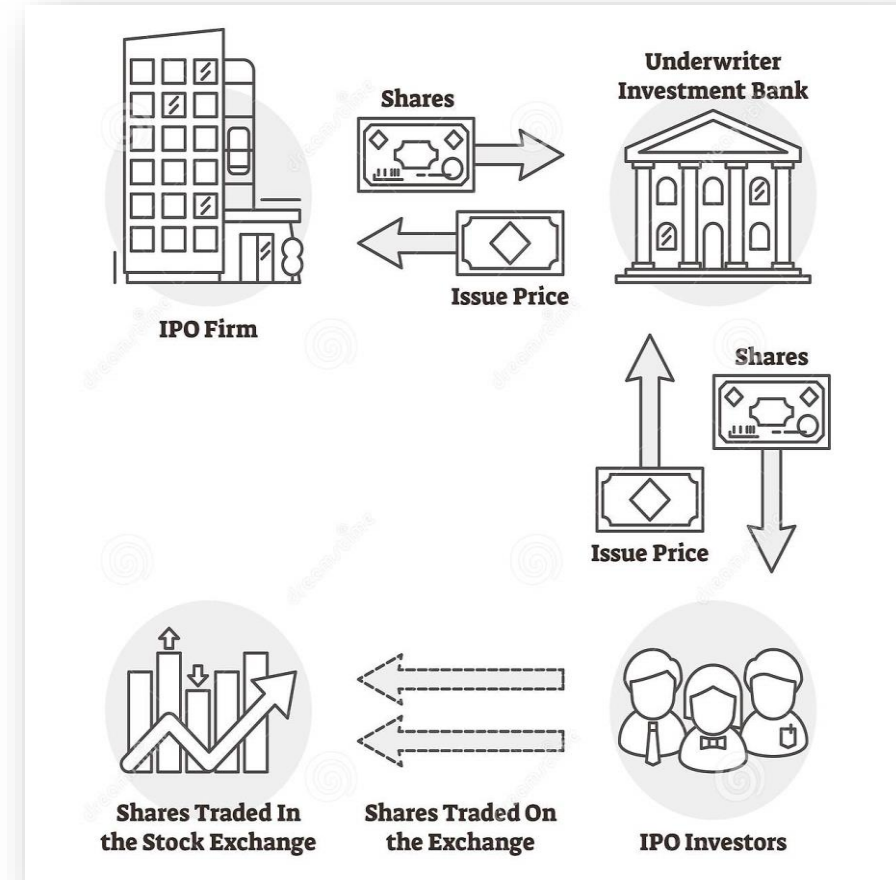


ASSET CLASSES AND RETURNS

Equity

How shares become Equity (common stocks): IPO

- Privately held companies (e.g., start-ups) have shares that are split between owners or partners
- When this company wants to raise capital from the public it will approach an investment bank to underwrite the offering (IPO)
- As the underwriter, the investment bank can choose 1 of 2 paths:
 - Best Efforts Commitment** - simply facilitates the sale to the public.
 - Firm Commitment Underwriting** – it buys all shares to resell them.
- IPO requires the company to also
 - develop and deliver offer documentation, including a prospectus.
 - use the services of other entities like accounting and law firms, (collectively, the selling syndicate)
- Post IPO, these shares are traded on the Stock Exchange as common stocks



Ways to buy/sell Common Stock

- Equity are traded in four types of securities markets
- Retail investors are typically only involved with the first two.

PRIMARY MARKET

- Where all initial public offerings (or IPOs) occur.
- These are stocks and bonds that are being offered to the public for the first time.
- When buying in this market, the investor is buying the security directly from the issuer.

SECONDARY MARKET

- The secondary market is where investors trade existing securities.
- Transactions in this market normally happen through one of the stock exchanges
- In these transactions, the seller of the securities, rather than the issuing company, receives the proceeds.

THIRD MARKET

- The third market is used by institutional investors and broker-dealers who are not exchange members, to trade large blocks of exchange-listed securities outside of exchanges, known as the “OTC” or “over-the-counter” market.

FOURTH MARKET

- The fourth market is also for institutional investors.
- Here, institutions buy and sell securities directly with one another through private computer networks.
- INSTINET (Institutional Networks Corporation), owned by Nomura Holdings in Tokyo, is an example of one of these networks.

Order types for common stock

- Common order types when buying or selling stocks

Market order

- Buy (or sell) now at current price, e.g., 150

Limit order

- Buy (or sell) only at specified price, e.g. buy at 140

Good till cancelled

- remains active until it is fulfilled or manually cancelled

Stop loss

- Execute as a **market order** if a specified price occurs (e.g., sell at market price if price drops to 125)

Stop limit

- Execute as a **limit order** if a specified price occurs (e.g., sell only at 125 if price drops to 125)

- Advanced strategies

Margin buying

Short sale

Stockholder rights (on owning 1 or more shares)

DIVIDENDS RIGHTS

- receive declared dividends in proportion to the number of shares of stock they own.

VOTING RIGHTS

- elect board members and vote during annual meeting and on all important decisions, e.g., stock splits etc

PREEMPTIVE RIGHTS

- right to purchase any newly issued shares of stock to maintain their ownership percentage before they're offered to the public.

OTHER RIGHTS

- Right to receive annual reports and know about the other stockholders.
- Right to claim during liquidation, and transfer ownership

STATUTORY VOTING

entitles the shareholder to cast one vote per share of stock for each position on the board of directors.

CUMILATIVE VOTING

entitles the shareholders to cast their votes in any way they want.

A **proxy** is a written power of attorney authorizing someone to vote on behalf of, and on the instructions of, a shareholder, when the shareholders will not be attending an annual meeting.

1. Capital appreciation

- The **increase in share price** over time
- **Realized on sale** of share
- Share price reflects **market perception**, not necessarily the **intrinsic value (true value)** of the share
- As a result, share can be **overvalued** or **undervalued** compared to the intrinsic value

2. Dividend income

- **Profitable and well-established companies issue dividends** to their shareholders at certain intervals
- Dividends may be **paid in cash or stock**
 - e.g., 10% dividend given 1 share for every 10 shares owned. Market adjusts such splits by reducing share price by the same amount.
 - Dividend payout ratio = total dividend payable / total profit

Equity return & valuation measures

Return

$$r = \frac{D_1}{P} + g$$

Where D1 = dividend+ anticipated growth,
g=appreciation, P = stock price

Dividend
payout ratio

Amount of dividend earmarked by company as a % of net profit, e.g., 30,000 total dividend to be paid out of total profit 100,000 = 30% dividend payout ratio

Stock split /
reverse split

Reduces price of stocks (or reverse split increases price)
E.g. 2 for 1 split will see stock price drop from Rs 100 to Rs 50

Stock value
(DDM)

$$V = \frac{D_0(1+g)}{r-g} \text{ or } V = \frac{D_1}{r-g}$$

Where r = expected return, g = anticipated dividend growth and V = Intrinsic value

Other Equity valuation measures

Price to sales	$P/S = \frac{\text{Stock price}}{\text{Sales per share.}}$	Where sales per share is company sales divided by number of outstanding shares Compared to industry PS ratio to identify over/under valued
Price to book	$P/B = \frac{\text{Stock price}}{\text{Book value per share}}$	Where book value is equity value divide by number of outstanding shares Compared to industry P/B values
Price to earnings	$P/E = \frac{\text{Stock price}}{\text{Earning per share}}$	Where Earnings per share is company profits for last 4 quarters (trailing), or projected for next year (forward) divided by number of outstanding shares
PE to growth	$PEG = \frac{\text{Price to Earnings}}{\text{EPS growth.}}$	Where PE ratio is divided by earnings growth rate, e.g., 5%

Fixed Income

Types of Fixed Income Securities

- Securities that provide a fixed income
- Involve lower risk than equity, and pay a modest return, usually in the 4-8% range
- Can be bought at IPO from underwriter, or through OTC markets

1. Cash & Cash equivalent

- Cash or Certificates of Deposit (CD) or money market funds (i.e., short term MF's)
- All have low return low risk
- used for emergency planning or as liquid reserves

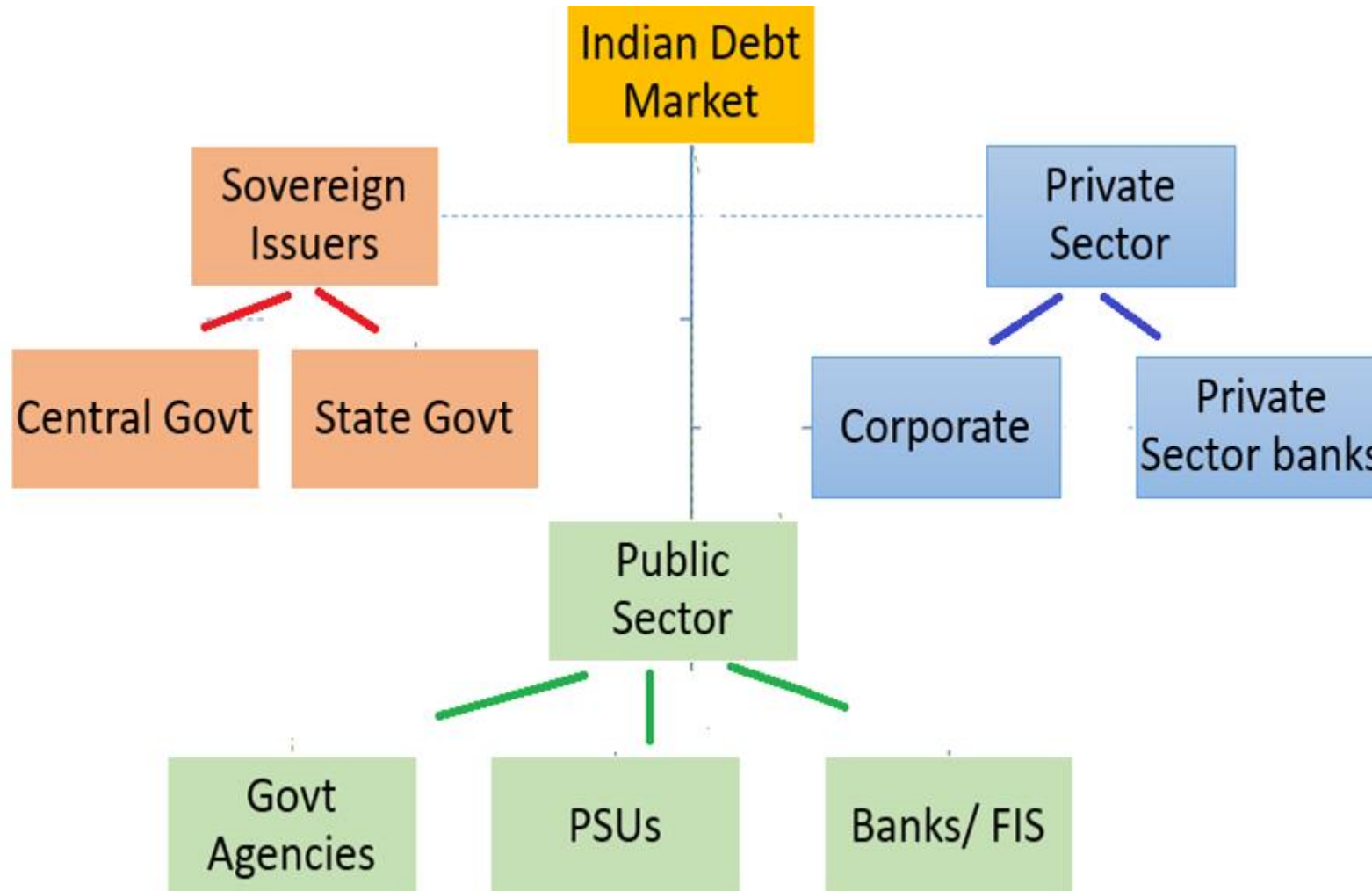
2. Bonds

- a securitized loan from an investor to a borrower (E.g., government, municipality, or company)
- Slightly higher return and risk compared to Cash equivalents
- Used to protect principle while earning modest returns

3. Preferred Stock

- pays a fixed dividend, i.e., provide a fixed income (which is paid ahead of common stock dividends)
- Unpaid dividends are cumulative
- Risks relate to interest-rate and purchasing-power

Bond issuers and types



Types of bonds

- Notes or bills
- **Convertible** bond
- **Put/Call** bond
- **Mortgage-backed / Asset-backed** bond
- **Zero-coupon** rate
- **Floating** rate
- **Inflation-linked**

Bonds: Key terms

- **Coupon rate (coupon)** is the annual interest rate paid by the bond issuer. Can be fixed / variable coupon (e.g., London Interbank Offered Rate -**LIBOR**)
- **Par value:** Issued at par value and traded (bought and sold) in the over-the-counter market or exchanges
- **Higher quality / Lower quality / Junk bonds:** Terms relating to the risk (& associated return) of a bond
- **Trading at discount / Premium:** When the coupon rate is lower / higher than current rates on new bonds
- **Terms of issue,** includes the bond's coupon rate, maturity date, and other conditions that may be relevant.
- **Call provision** allows the bond issuer to “call in” and redeem the bond at a predetermined price
- **Call premium:** When the call price is higher than the face value, the difference is referred to as the call premium.
- **Call protection:** Bond is said to have protection in the years between the issue of a bond and its call date.
- **Sinking fund:** a series of periodic payments made by the issuer to retire a bond issue in tranches

Return (& risks) of Fixed Income Securities

1. Interest income

- Received at fixed intervals or sometimes on maturity
- Can be traded during their tenure to generate profit / loss based on market price relative to bond face value (i.e., issue terms)

2. Principal recovery

- Recovered on maturity
- Some high-risk bonds can default on principal repayment
- CDS (Credit default swap) = institutions pay premiums to a CDS seller (i.e., insurance against default by bond issuer)

Bonds may be subject to the following risks:

ENDOGENOUS

INTEREST-RATE

MARKET

SOCIAL POLITICAL

CURRENCY

INFLATION

CALL

LIQUIDITY

FINANCIAL

CREDIT

Bond valuation and return measures

Bond present value / price

A bond's **present value** changes as **current market interest rates change**, E.g., if current interest rates > coupon rates, **bond is below par** (i.e., since bond PV is < Par value) and **is above par** (PV > Par value) if interest rates < than coupon rate

Nominal yield

$$= \frac{\text{Annual interest}}{\text{Bond's par value}}$$

the annual interest divided by the bond's par value

Current yield

$$= \frac{\text{Annual interest}}{\text{Current bond price}}$$

annual interest divided by the bond's current price

Yield to maturity (YTM)

$$= \frac{\left(\text{Annual interest} + \left(\frac{\text{Par value} - \text{Current bond price}}{\text{Number of years to maturity}} \right) \right)}{\frac{\text{Par value} + \text{Current bond price}}{2}}$$

total return earned on a bond if it is held to maturity.

Yield to call (YTC)

$$= \frac{\left(\text{Annual interest} + \left(\frac{\text{Call price} - \text{Current bond price}}{\text{Number of years to first call date}} \right) \right)}{\frac{\text{Call price} + \text{Current bond price}}{2}}$$

total return earned on a bond if it is held to the time it is called

Tax-free bonds

- An investor needs to be able to compare returns on a tax-free bond to a regular bond.
- E.g., is a 3% tax free bond better or worse than a 5% taxable bond?

Duration (risk)

- Measures the **interest rate risk** of fixed income securities
- Used by investors **who intend to sell before maturity**
- E.g., if interest rates rises 1%, a bond with a duration of six would decrease in value about 6%.

Yield Curve

- shows whether it is better to hold short-, medium- or long- term bonds
- **Positive** curve: Pays better in Long term
- **Negative** curve: Pays better in short term
- **Flat**: no difference

Preferred Stocks valuation

- Preferred stock generally pays a fixed dividend, thereby providing a fixed income to the investor.
- The major risks of preferred stock, then, are interest-rate risk and purchasing-power risk.
- If interest rates increase or decrease, dividend yields on preferred stock tend to follow as investors compare them to yields on other fixed-income securities, primarily bonds.
- The valuation formula for preferred stock is: $V = \frac{D}{K}$ or $V = \frac{D}{r}$
- Where:
 - V = the preferred stock value
 - D = the annual dividend
 - k = the market yield for similar quality preferred stocks
 - r = the required rate of return (zero growth model)

PREFERRED STOCK VALUATION : EXAMPLE 1

Maria wants to know the value of her preferred stock. It has an annual dividend of \$5 per share and her required rate of return is 5%. Using the zero-growth model, what is the current value of her stock?

Solution:

$$V = \frac{D}{r}$$

$$V = \$5/.05$$

Where:

V = the preferred stock value

D = the annual dividend

The current value is \$100

Pooled Products

Mutual Funds, ETF's, ETN's, UIT's, Managed Accounts

1. Mutual Funds

- Most popular category of pooled products
- Investor's funds are pooled together, and a professional investment advisor is hired to invest that money into several different securities
- Three broad categories of MF's are available

Growth

- Aimed at **appreciation of capital**
- Equity is primary composition

Income

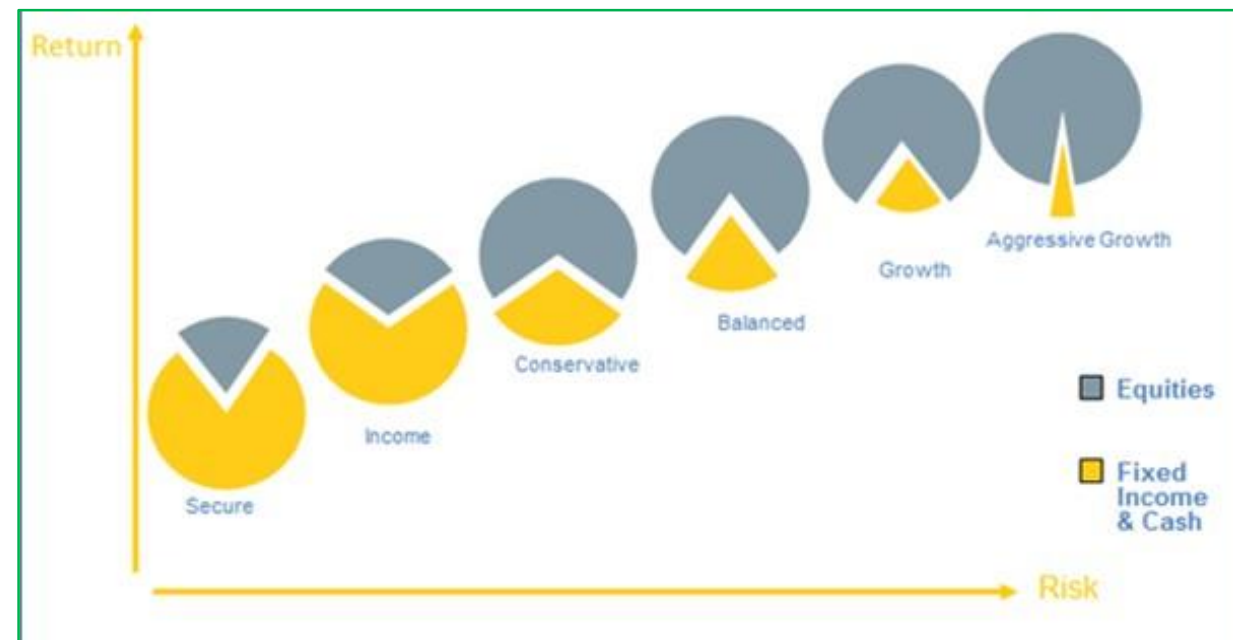
- Aimed at **regular pay-out to participants**
- Fixed income securities are primary composition

Preservation

- Aimed at **preventing any loss to capital invested**
- 100% of fund is cash and fixed income

Mutual Funds: Composition examples

- Each individual fund will have a target mix of low risk and higher risk instruments.
- These are usually split as fixed income / cash (safer) and equities (higher return/risk)



Mutual Funds	SECURE		INCOME		CONSERVATIVE		BALANCED		GROWTH		AGGRESSIV E GROWTH	
	Range	Target Mix	Range	Target Mix	Range	Target Mix	Range	Target Mix	Range	Target Mix	Range	Target Mix
Asset Allocation												
Cash	0%-35%	20%	0%-35%	5%	0%-25%	5%	0%-20%	5%	0%-20%	5%	0%-20%	5%
Fixed Income	65%-100%	80%	65%-95%	80%	45%-75%	60%	25%-55%	35%	5%-35%	20%	0%	0%
Equities	0%	0%	5%-25%	15%	15%-55%	35%	35%-75%	60%	55%-95%	75%	80%-100%	95%
Total		100%		100%		100%		100%		100%		100%

- MF's have both advantages and disadvantages

Advantages

- Professionally managed
- Diversified
- Low entry cost / easy to enter

Disadvantage

- Tax (depends on country)
- Herd behaviour
- Pricing of MF shares
- Inbuilt costs

2. Exchange Traded Funds (ETFs)

- An ETF will **track one index and replicate its returns**, e.g. gold, stock, commodity etc
- ETF's can be constructed in two ways
 - Include **every security** in the index, a strategy known as **full replication**.
 - Include a **representative sample** of the index, called **sampling replication**

Advantages

- Cost Efficiency: ETFs have significantly lower expenses than mutual funds
- Tax Efficiency: ETF owners control the timing of their purchases and sales
- Availability of Option Contracts

Disadvantages

- NAV and Tracking Error: Do not always trade exactly at their net asset value, leading to performance that differs from the index.
 - The primary cause of tracking error in an ETF relates to how it is constructed (replicant / sampling as above)

3. Exchange-Traded Notes (ETNs)

- ETN's are loan instrument issued by a financial entity, such as a bank. They do not have any underlying securities as they are linked to a particular stock index.
- It comes with a set maturity period, usually from 10 to 30 years.
- ETNs are linked to a stock index, thus do not possess any voting rights, nor pay a fixed rate of interest.
- The gains or losses of the investor are derived from the performance of an asset class or index that it tracks.
- ETNs are most commonly used to track commodity and currency exchange markets, where mutual funds and ETFs aren't typically available.
- ETNs are tied to the **creditworthiness** of the issuer. This influence investors to buy.

4. Unit Investment Trusts (UITs)

- UITs are **registered investment companies** that pool investor money, but unlike the others, UIT holdings do not change.
- For this reason, they are sometimes also referred to as fixed trusts or defined portfolios.
- UITs can contain different types of securities (e.g. stocks, preferred stocks, bonds, REITs, etc.) that are professionally selected but not subsequently traded.
- Like mutual funds, UITs are priced daily, after markets close.

5. Managed Accounts

- Sometimes referred to as separately managed accounts (SMAs) or privately managed accounts (PMAs),
- These are individual (or joint) accounts that are managed by a professional investment manager, directly for the account owner.
- Unlike a mutual fund investors in managed accounts own the individual securities directly.

Real Estate

- Real estate investing involves investing in real estate for profit.
- Investing in Real Estate can be in three ways

Outright purchase

- **Purchase** of residential, commercial or land
- **Generates income** from sale, rent, agricultural produce

REIT (RE investment Trust)

- **Works like a Mutual Fund**
 1. **Equity** REIT: based on buying property
 2. **Mortgage** REIT: based on funding RE projects
 3. **Hybrid** REIT: mix of equity and mortgage

Limited Partnership

- a **group of investors pool their money** to form a corpus
- one partner manages this corpus
- partners share income, capital gains, losses, and deductions

- Intrinsic value of Real Estate can be valued in two ways

NOI Capitalization method

- Estimates value based on **income generation of property**
- **Formula:** Property Value = **NOI/Cap Rate**
- E.g., Net operating Income is 5 lacs and annual appreciated at 8%, hence value = $5L/8\% = 62.5L$

Gross income multiplier

- Estimates value based on **sale of comparable units**
- GIM = sale price (of comparable unit) as a ratio of gross annual revenue, e.g., sold at 60L, and annual rent = 6 lacs, so $GIM=10$
 - **Does not consider expenses** incurred in maintaining property

Commodities, Futures & Options

Commodities (Futures)

- Commodities are normally hard assets (e.g., energy, grains,, metals, etc.)
- **Commodity suppliers (e.g., farmers) / buyers (e.g., wholesalers)** want to de-risk future price or supply fluctuations, so they enter a **futures contract**
- These **futures contracts**
 - Usually eliminate need to take physical delivery of actual commodity, e.g., corn, timber, coal etc
 - Are priced at **current spot rates**
 - Specify a **future date, rate and quantity** .
- **Speculators** are investors who look to make a profit through futures



Options

- An **alternative to buying an asset** (e.g., 100 equity shares)
- By **paying a small premium** (e.g., 5% of the underlying asset) **you get an Option**
- Options can be of two types as below

CALL OPTION	PUT OPTION
Right to buy (i.e., call for delivery of) the underlying asset	Right to sell (i.e., put up for sale) the underlying asset

- | | |
|--|---|
| <ul style="list-style-type: none">• Buyer can, but is under no obligation, to buy the underlying assets• Seller is obliged to sell the underlying, if exercised.• E.g., Buyer has option to buy 100 shares of XYZ at Rs 50. Seller must sell these shares if exercised. | <ul style="list-style-type: none">• Buyer can, but is under no obligation, to sell the underlying assets• Seller is obliged to buy the underlying, if exercised.• E.g., Buyer has option to sell 100 shares of XYZ at Rs 40. Seller must buy these shares if exercise. |
|--|---|

How Options work

- Options can be on any asset, e.g., equity, debt, currency, etc

Call

Buyers

Expects price to rise, so they can **make a profit by buying asset below market price**

Sellers

Expect price to fall, so they can **profit from the option premium** (since buyer will not exercise)

Put

Buyers

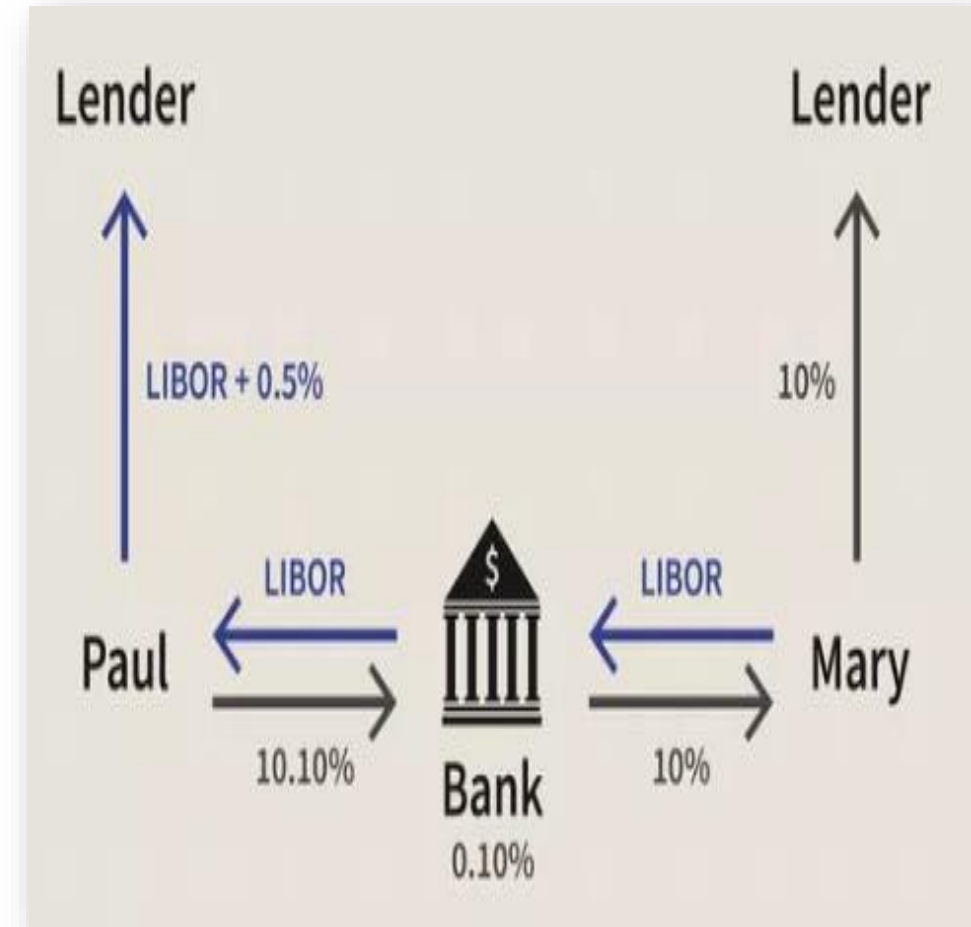
Expect price to fall, so they can make a **profit by selling asset above market price**

Sellers

Expect price to rise, so that they make a **profit from the option premium** (since buyer will not exercise)

SWAPS

- Swaps are option that typically swap a **fixed-interest income** stream from one party for a **floating rate** from another party.
- The floating rate payments are typically based on the London Interbank Offered Rate, or LIBOR.
- Banks often initiate swaps and offer them to corporations, another bank or investors.
- The types of swap are **interest rate swap**, **currency swap**, **commodity swap** and **credit default swap**.



Structured / Market Linked

Structured Products/Market Linked Securities

- investment vehicles whose value is derived from a reference asset, market measure or investment strategy, e.g., single equity or debt securities, indexes, commodities, interest rates and/or foreign currencies or a mix of these
- Structured products typically have two components: a fixed income security (pay interest at a specified rate and interval) and a derivative, which is often an option (payment at maturity).

Growth

- Appreciation focussed
- generally, do not offer a full return of principal (e.g., 40% downside protection)

Yield

- provide higher current yields than what could typically be achieved by investing directly

Market-linked

- Provides market-linked returns
- Usually returns most of the principal

Access

- Provide access to global securities or sectors otherwise difficult to access by investors

Alternative investments

- Are aimed at HNI customers and institutional investors
- Can be of following types

Venture capital

- Buying **minority stake** in a small / **innovative start-up**
- Which **lacks other ways of raising funds** (e.g., via IPO or bank loans)

Private equity

- Buying **majority stake** in a larger, more **established company**,
- which is **facing financial or other business issues**

Hedge fund

- A **form of pooled products**, where investments are managed by a fund manager
- More risk-taking and less regulated than mutual funds

Forex

- Invest in **foreign exchange and currency markets globally**
- Dominated by banks and financial institutions

Collectibles

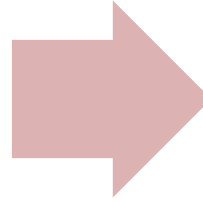
Real Assets and Collectibles

- Real assets includes antiques, art, figurines, coins, stamps, gemstones, comic books, and just about anything else that someone might want to hold for possible future sale
- Collectibles are among the riskiest of investments and requires the investor to have significant knowledge about the item since
 1. they produce no income.
 2. Many have costs of ownership (insurance, storage, security).
 3. No active market exists for most collectibles which makes it difficult for the investors able to buy or sell an asset easily, quickly, or perhaps at all.
- When trying to place a value on a collectible, consideration must be given to the item's condition, age and quality, the general availability of similar items, and the market's demand for the item.
- It should be clear that collectibles may be enjoyable to hold, but most people should not consider them as potential investments.

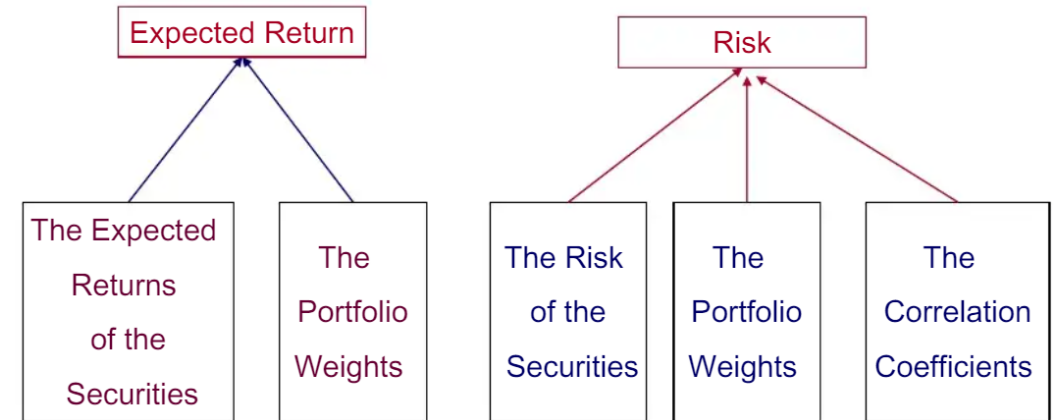


Risk & return: Part 1 (Risk)

Individual assets



Portfolio of assets



What is Risk

- **Risk = likelihood** of a loss, e.g., reduction in potential earnings, loss of capital etc.
- **Total Investment Risk = Systematic Risk + Non-systematic Risk**

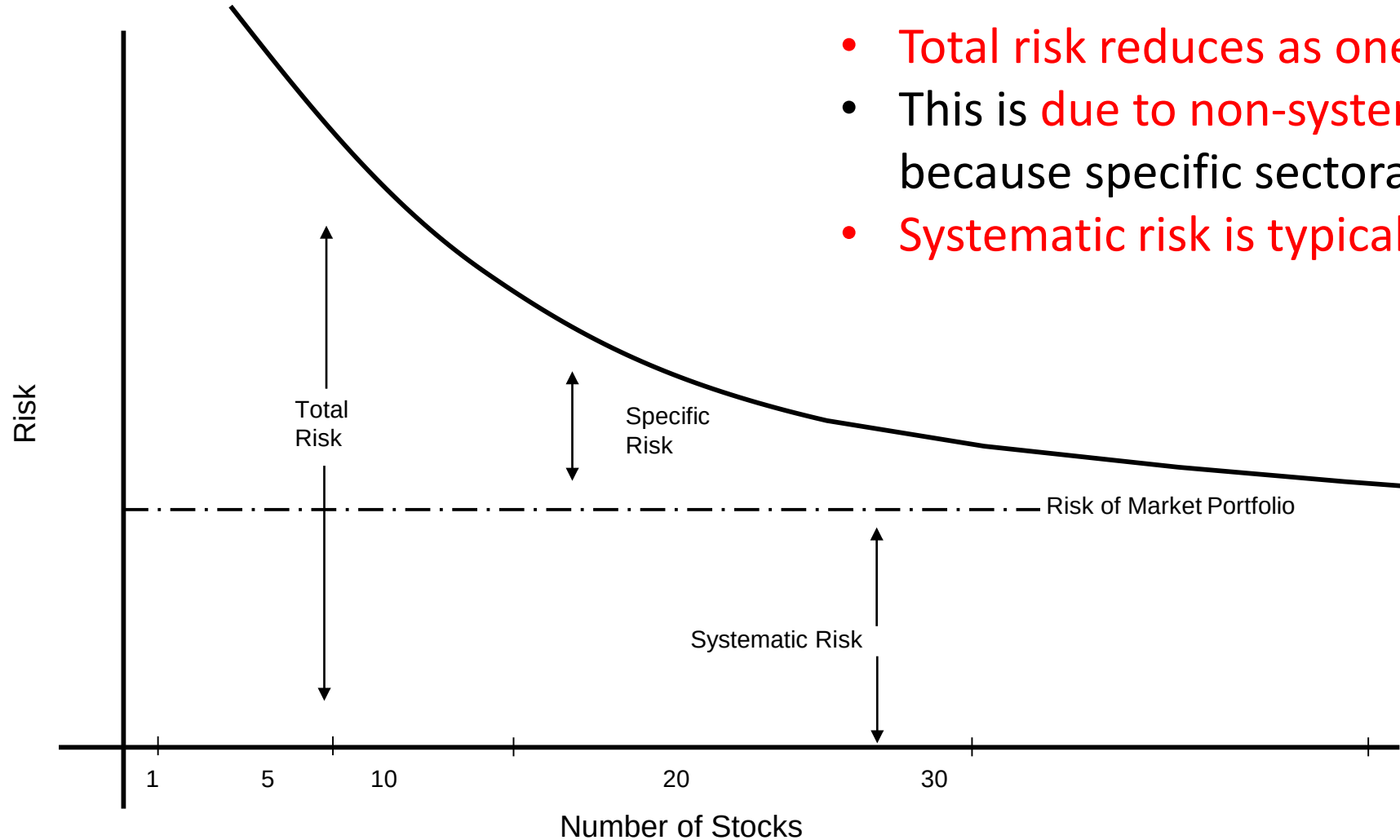
Systematic (or market) risk

- Based on **overall economy or securities markets** (e.g., interest rates, bull or bear market, inflation, geopolitical)
- **Difficult to reduce**, and comprise of
 1. **Endogenous** (or internal) risks within economy or financial system
 2. **Exogenous** (or external) risks arising from sectoral news or events

Non-systematic risk

- Comes from **investing in a specific company or sector** (e.g., poor leadership, losses, debt burden)
- **Can be reduced via creating a diversified portfolio**

Total Risk and portfolio diversification



- Total risk reduces as one adds more securities
- This is due to non-systematic risk reduction (i.e., because specific sectoral risks offset each other)
- Systematic risk is typically constant

Total risk

- Standard deviation

Non-systematic risk

- Covariance
- Correlation

Systematic risk

- Coefficient of determination / Beta

Total risk: Calculating SD

Find the **mean** for your data

Mean of {1,2,3,4,5} = 3



Find the **deviation** of each data point from the mean

(1-3), (2-3), (3-3), (4-3), (5-3)



Square each of these deviations

$(-2)^2 = 4$
 $(-1)^2 = 1$
 $(0)^2 = 0$
 $(1)^2 = 1$
 $(2)^2 = 4$



Sum these values and **divide** by **n-1**
(= Variance)

Sum = 10, divide by 5-1 = 4
= **2.5** (Variance)



Find **square root** of variance
(= Std Deviation)

Sq. root of 2.5
= **1.58** (SD)



$$\sigma = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

Where $\bar{X}$ = Mean
and $\sum$ = Sum

Graph of returns over time: Skewness

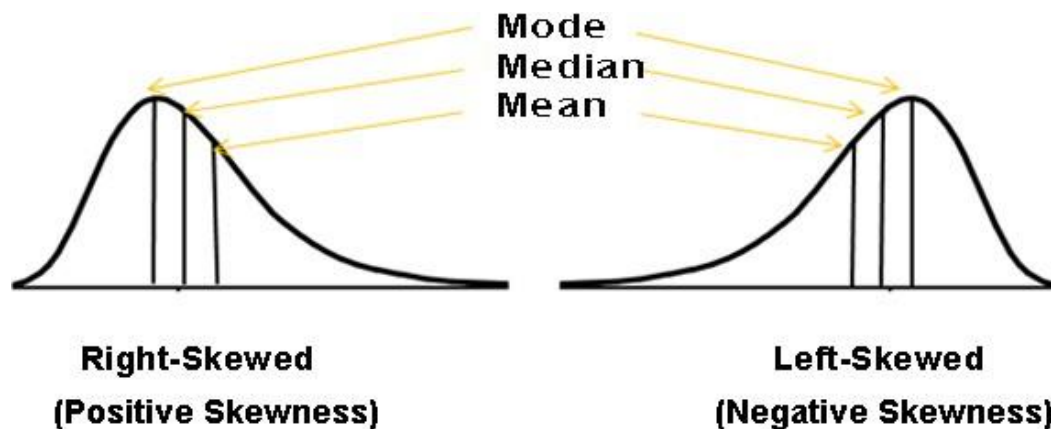
Distributions which are **not normal** are called Skewed

Skew can be either left-aligned or right-aligned

Right-skewed implies most data falls on the right (i.e. centre is after the peak)

Left-skewed implies most data falls on the left (i.e. centre is before the peak) :

Note: left or right is based on which side is central tendency NOT on which side the peak is



Graph of returns over time: Kurtosis

Kurtosis means **bulge** in Greek and measures the peakiness

This represents the spread of a stock

Lots of outliers = High peak = High risk

Normal tail = normal peak

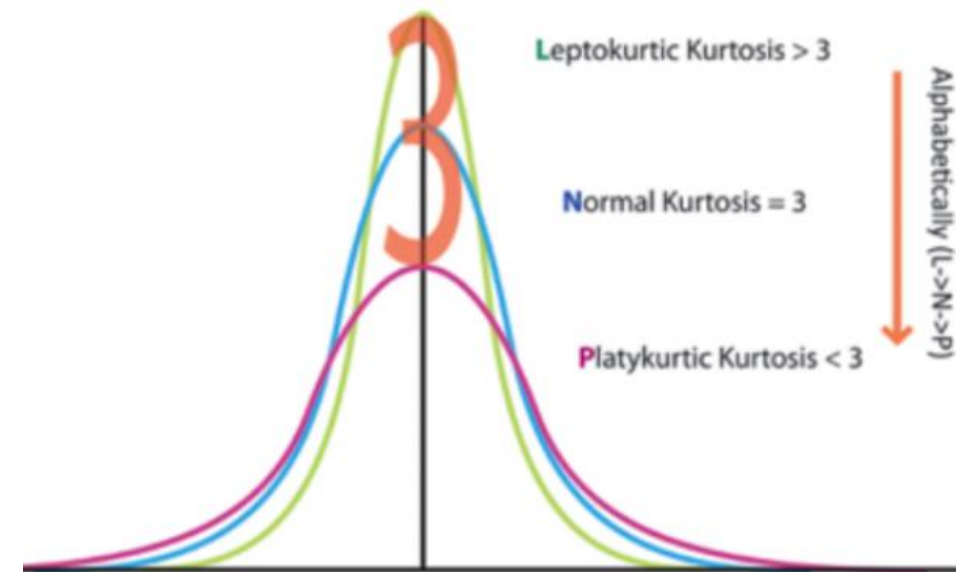
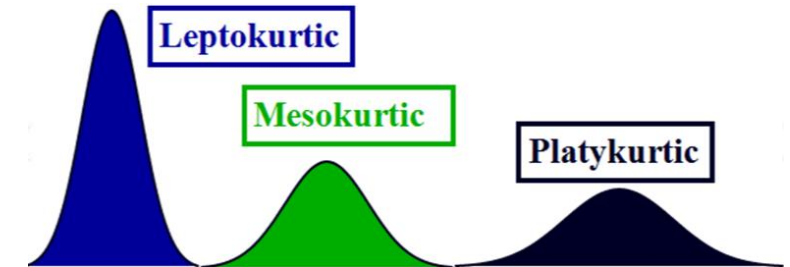
Few outliers = low peak = Low risk

It's value is calculated as K

Leptokurtic ($K > 3$), positive Kurtosis

Mesokurtic / Normal distribution ($K = 3$)

Platykurtic ($K < 3$), negative Kurtosis



- It identifies the likelihood of two variables (investment returns, in our context) moving together, over time.
- Covariance formula =

$$cov_{xy} = \sum_{i=1}^N \frac{(x_i - \underline{x})(y_i - \underline{y})}{n - 1}$$

- Cov= **Product of individual variances**, divided by **period-1**
- Here it is - $\frac{3}{4} = -0.75$

1 Period n	2 Return Stock A % x_i	3 Deviation from Average $(x_i - \underline{x})$	4 Return Stock Z % y_i	5 Deviation from Average $(y_i - \underline{y})$	3 x 5 Product of Deviations $(x_i - \underline{x})(y_i - \underline{y})$
1	9	-1.00	14	3.00	-3.00
2	10	0.00	11	0.00	0.00
3	11	1.00	9	-2.00	-2.00
4	8	-2.00	10	-1.00	2.00
5	12	2.00	11	0.00	0.00
Average	$\underline{x} = 10$		$\underline{y} = 11$		Sum $\Sigma = -3$

We used column 6 and divided by n-1

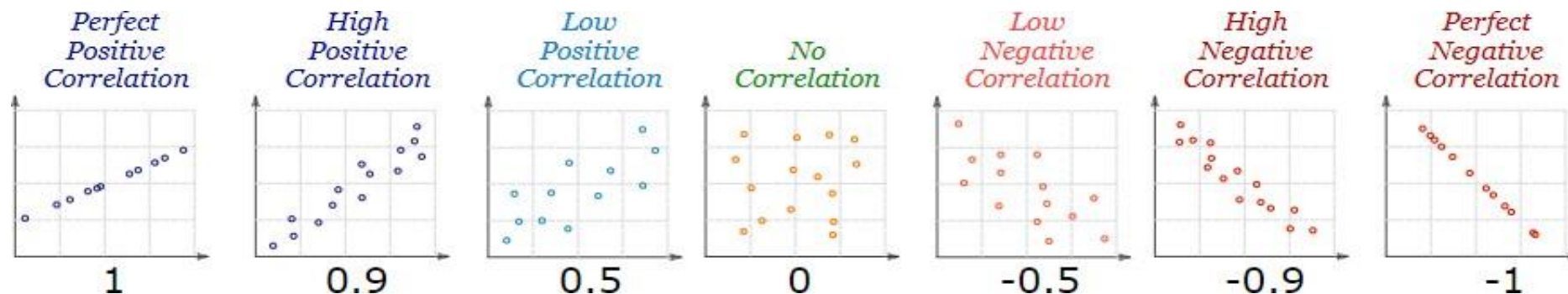
Non-systematic: Covariance and Correlation Coefficient

➤ Covariance can be +ve or -ve. It tells us the nature of the relationship.

➤ Correlation coefficient tells us the strength of this covariance

➤ **Formula** is Cor or R (used interchangeably) $Cor_{xy} = \frac{cov(X,Y)}{\sigma_x\sigma_y}$ or $R_{xy} = \frac{cov(X,Y)}{\sigma_x\sigma_y}$

➤ The correlation coefficient between two variables identifies their degree of correlation (i.e., relationship), using a scale of **+1 (perfectly positive)** and **-1 (perfectly negative)**.



Summary: Covariance and correlation

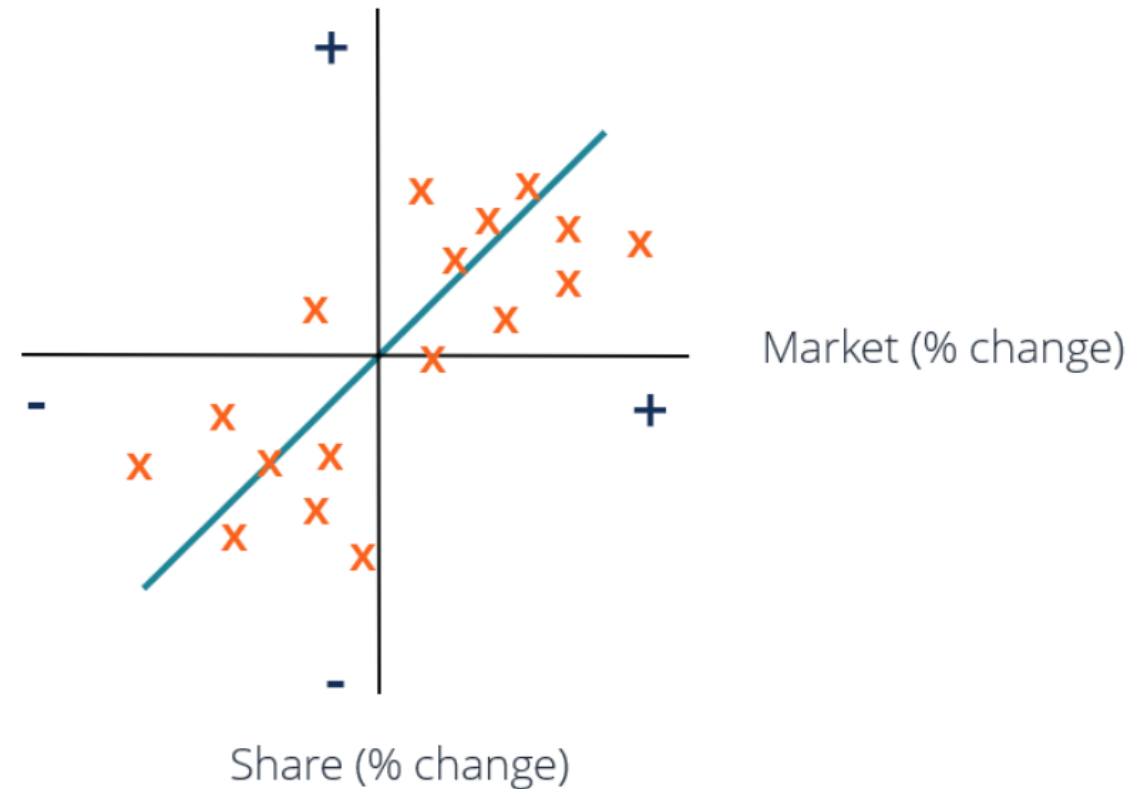
- **Covariance** and **correlation** are two terms that are **exactly opposite** to each other. However, they both are used in risk analysis.
- Covariance shows us how the two variables vary, i.e. co-variance
- Correlation shows us the relationship, i.e. co-related
- They are dependent on each other as **Covariance = Correlation x SD_1 x SD_2**

	Covariance	Correlation
Meaning	Covariance is an indicator of how two random variables are dependent on each other. A higher number denotes higher dependency.	Correlation indicates how strongly these two variables are related , provided other conditions are constant. The maximum value is +1, representing a perfect dependent relationship.
Inter-relationship	Covariance is used to deduce correlation	a measure of covariance on a standard scale. It is deduced by dividing the calculated covariance by standard deviation.
Expected values	The value of covariance lies in the range of $-\infty$ and $+\infty$.	Correlation is limited to values between the range -1 and +1.

- Systematic risk is undiversifiable
- It is valuable to be able to determine how much of a portfolio's (or asset's) risk is undiversifiable.
- This measure is called the coefficient of determination, R^2 , or R-squared.
- **Formula:** The correlation coefficient is squared, i.e. if $R = 0.3$, then R^2 is 0.09
 - As an example, correlation coefficient between Portfolio A and the FTSE 100 is 0.30. Squaring the correlation coefficient, the result is 0.090.
 - This means that 9.00% of Portfolio A's price movement is directly associated to the movement of the FTSE 100 (i.e., systematic risk).

- Beta is also used to measure systematic risk.
- If an asset or portfolio has a beta of 1.00 against its benchmark(s), and the market goes up 10.00%, the asset can be expected to also rise 10.00%.
- As a rule, clients who identify themselves as being **conservative** will want to focus on assets that have a **beta below 1.0**, or at least very close to 1.0 on the upside.

Beta = slope of the line



The formula to calculate an asset's beta is to divide the covariance of the asset and the market by the variance of the market.

$$\beta = \frac{cov_{im}}{\sigma^2_m}$$

This becomes the commonly used formula for Beta:

$$\beta = R_{im} \times \frac{\sigma_i}{\sigma_m}$$

Where:

β = beta

R_{im} = correlation coefficient between market and asset

σ_i = standard deviation of the asset

σ_m = standard deviation of the market

Beta for a portfolio

Portfolio beta uses a weighted average approach.
To calculate it, we need individual stock betas
This can be extended to multi-stock portfolios

Asset	Balance	Beta	Portfolio Weight	Beta x Weight
A	70,000.00	0.80	70%	$0.80 \times 0.70 = 0.56$
Z	30,000.00	1.50	30%	$1.50 \times 0.30 = \underline{0.45}$
				1.01

Beta: Practical

N	Date	GE		1	Variance	S&P 500		2	Variance	Covariance	1 x 2
		Closing price	Rate of return %	R - ER	(R-ER) ²	Closing price	Rate of return %	R - ER	(R-ER) ²		(R-ER)*(R-ER)
0	Dec-03	30.98				1,111.92					
1	Jan-04	33.63	8.55	7.11	50.50	1,131.13	1.73	0.99	0.97		7.02
2	Feb-04	32.52	-3.30	-4.75	22.55	1,144.94	1.22	0.48	0.23		-2.28
3	Mar-04	30.52	-6.15	-7.60	57.72	1,126.21	-1.64	-2.38	5.65		18.05
4	Apr-04	29.95	-1.87	-3.32	10.99	1,107.30	-1.68	-2.42	5.85		8.02
5	May-04	31.12	3.91	2.46	6.05	1,120.68	1.21	0.47	0.22		1.15
6	Jun-04	32.4	4.11	2.67	7.11	1,140.84	1.80	1.06	1.12		2.82
7	Jul-04	33.25	2.62	1.18	1.38	1,101.72	-3.43	-4.17	17.38		-4.90
8	Aug-04	32.79	-1.38	-2.83	8.01	1,104.24	0.23	-0.51	0.26		1.45
9	Sep-04	33.58	2.41	0.96	0.92	1,114.58	0.94	0.20	0.04		0.19
10	Oct-04	34.12	1.61	0.16	0.03	1,130.20	1.40	0.66	0.44		0.11
11	Nov-04	35.36	3.63	2.19	4.78	1,173.82	3.86	3.12	9.73		6.82
12	Dec-04	36.5	3.22	1.78	3.16	1,211.92	3.25	2.51	6.28		4.45
			17.37		173	sum	8.88		48	sum	43
		Average				Expected Return					
		=	1.45	Variance	15.75	0.74		Variance	4.38	Covariance	3.90
		is in same units as data =		Std. Dev	3.97			Std. Dev	2.09	Beta	0.89
				Variance	15.75			Variance	4.38		
		portfolio	50%	Std. Dev	3.97	50%		Std. Dev	2.09		



Risk & return: Part 2 (Return)

1. Returns

Calculate returns

- Weighted, Time-weighted, IRR, HPR

Compare to goal

- Sharpe, Treynor, Jensen

Compare to market

- Market capitalization, Price-weighted, Equally-weighted

2. Buy/sell decisions

Fundamental
analysis

Technical
analysis

Contrarian
approach

1. Weighted Average Return

- Weighted average is universally used to describe multi-asset portfolios
- It is the average of the returns of each holding in the portfolio, adjusted by the percentage weighting of each holding in the portfolio.

Holding	Initial value	Annual return	Weight	Annual return x Weight
Fund A	\$40,000	10%	0.4	$10 \times 0.4 = 4.0\%$
Fund B	\$20,000	6%	0.2	$6 \times 0.2 = 1.2\%$
Fund C	\$30,000	8%	0.3	$8 \times 0.3 = 2.4\%$
Fund D	\$10,000	14%	0.1	$14 \times 0.1 = 1.4\%$
Total	\$100,000		1.000	9.00%

2. Time Weighted Return

- The geometric, or time-weighted, return is **most appropriately used for evaluating the performance of a portfolio over time.**

$$\text{GAR} = \sqrt[n]{(1 + r_1) * (1 + r_2) * \dots * (1 + r_n)} - 1$$

- r_1 r_2 etc are the returns in each year. This formula adjusts the invested amount at the end of each compounding period. E.g. \$100 → \$90 → 112.5

Investment return for the year	1	2	3	4
	-10%	25%	-15%	5%

- $\text{GAR} = \sqrt[4]{(1 - .10) \times (1 + 0.25) \times (1 - 0.15) \times (1 + 0.05)} - 1$
- In case returns are quarterly, use $[(1 + \text{Q1 return}) \times (1 + \text{Q2 return}) \times (1 + \text{Q3 return}) \times (1 + \text{Q4 return}) - 1.00] \times 100$

3. Dollar-Weighted Return / IRR

- the most appropriate measure of return to use with individual portfolios
- Also known as a money-weighted return or internal rate of return (IRR).
- It also allows for the **size and timing of cash flows into and out of an investment.**

Year	1	2	3	4
Starting balance	\$0	\$4,500.00	\$ 7,875.00	\$7,393.75
New investment at start of year	\$5,000.00	\$3,000.00	\$ 2,000.00	\$0
Net balance at the beginning of year	\$5,000.00	\$7,500.00	\$ 9,875.00	\$7,393.75
Investment return for the year	-10%	25%	-15%	20%
Investment gain (loss)	\$(500.00)	\$1,875.00	\$(1,481.25)	\$1,478.75
Withdrawal by investor at end of the year	\$0	(\$1,500)	(\$1,000)	\$0
Balance at the end of year	\$4,500.00	\$7,875.00	\$ 7,393.75	\$8,872.50

CASHFLOW

Y0= -5,000

Y1= -3000

Y2 = -500 (withdrawal 1,500 + new 2000)

Y4= 1000 (withdrawal 1000, no new)

Closing (Y4) = 8,872.50

IRR = 4.45531.

4. Holding period return

- This return calculates the **return over a holding period without annualizing returns**.
- It is simply the **difference between starting and ending value**, net of any interim income or payments made

- Formula: $HPR = (S + NI - P)/P$

Where:

S = sales price/proceeds

NI = net income/expense (+ dividends, - margin interest)

P = purchase price/cost

- *E.g. purchased at 100, received 17 as income and paid 3 as expenses and sold at 130 after 4 years. Hence $HPR = (130 + 14 - 100)/100 = 0.44$*

Comparison to goal (3 measures)

- Once we have calculated actual returns, our next task is to compare against goals

Sharpe ratio

- Relative measure
- Compares with similar investment or market index (r_f)
- Uses total risk (SD)
- Higher ratio is better

$$= \frac{r_p - r_f}{\sigma_p}$$

Treynor

- Relative measure
- Used for fully diversified portfolios, where only systematic risk (Beta) is expected
- Higher ratio is better

$$= \frac{r_p - r_f}{\beta_p}$$

Jensen (Alpha α)

- Absolute measure
- compares to expected return / benchmark (r_m)
- If α is +ve, portfolio outperformed goal

$$r_p - [r_f + (r_m - r_f)\beta]$$

Comparison to market (benchmark construction)

- Finally, the investors want to know how their investments against the market.
- A suitable market benchmark therefore needs to be defined basis 1 of the following

Weighted by Market Capitalization

- Weight of each security is its **market cap. divided by the sum of all the securities' market cap.**
- e.g. Security A's weight is 10% as its market cap is \$1million and sum of portfolio market caps is \$10million

Weighted by price

- Weight of each security is its **market price divided by the sum of all the securities' market prices.**
- e.g. Security A's weight is 5% as its price is \$100 and sum of all portfolio prices is \$2000

Equally weighted

- Weight of **each security is equal**
- E.g., Security A's weight is 10% as there are total of 10 securities in the portfolio

- Study of financial fundamentals to **predict the future performance of a company.**
- Results in an intrinsic value, that is compared to current stock price to take buy/sell/hold decisions

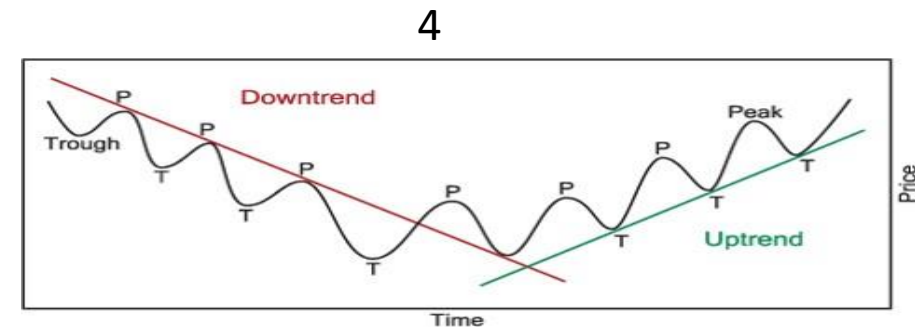
Top-down approach

- **Overall view** of
 - 1.Economy (i.e., business cycles, fiscal & monetary policy etc)
 - 2.Sector/Industry (i.e., growth, cyclical, defensive etc)
 - 3.Company (i.e., market share, Ratios, Leadership etc)
- Requires financial model to be built

Bottom-up approach

- **Company-specific view**, focusses on company's fundamentals, based on its financial statements and strategies
- Financial model is built to project future performance of the company
- Ignores economy and sector

- This is a different approach to Fundamental analysis. It focussed only on the analysis of the patterns and direction in price movement, and **assumes the market knows best**.
- Technicians decide on buy/hold/sell based on 4 questions
 1. What is the **overall trend**? i.e., are moving averages trending upwards or downwards
 2. Is there a **support level**? i.e., lower price limit, below which stock is bearish
 3. Is there a **resistance level**? i.e., upper price limit, beyond which stock is bullish
 4. What is the **momentum**, or trend? i.e., recent trends and trading volumes

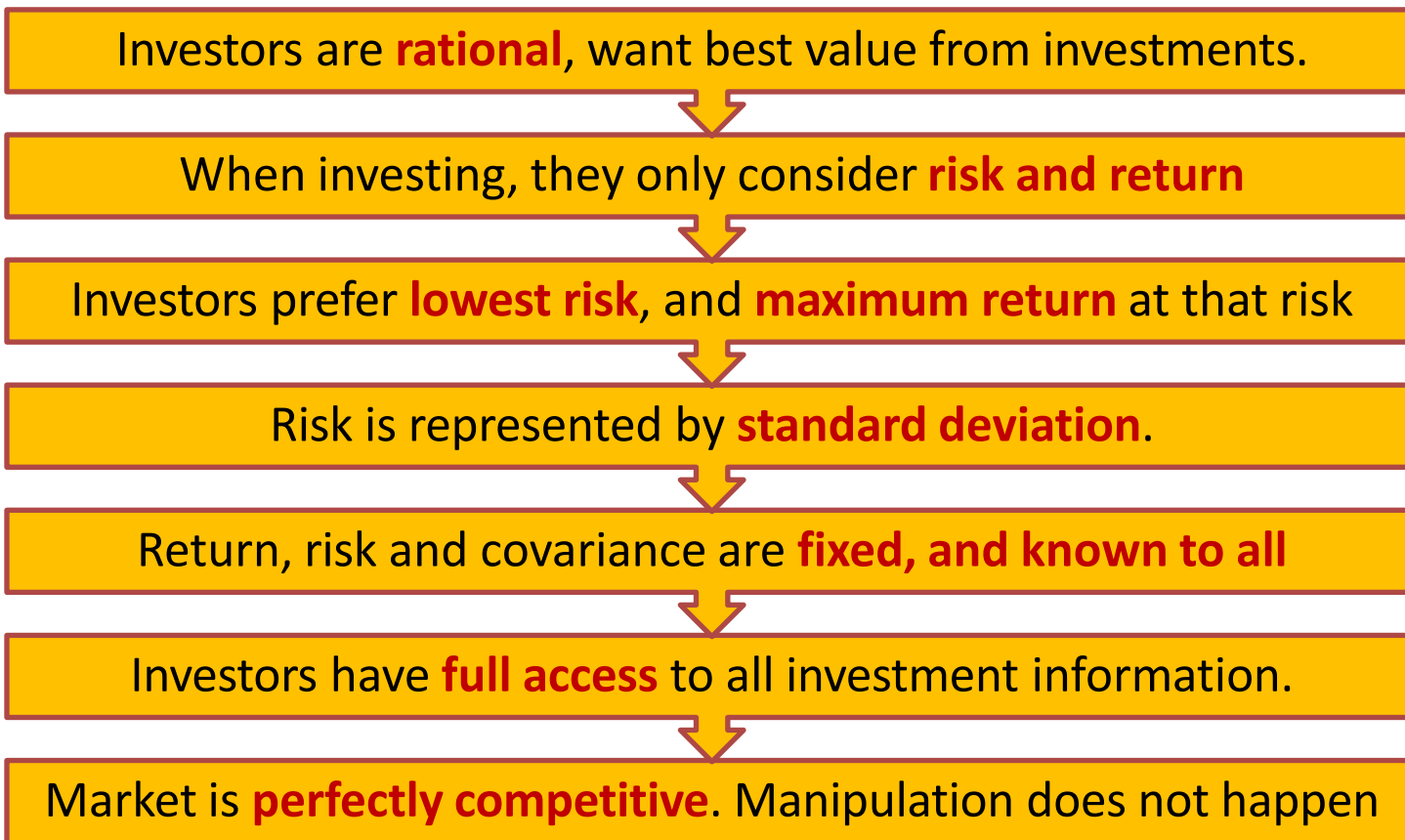


- Investors who identify themselves as contrarians believe that most investors are incorrect in their investment decisions
- They will take investment action that is opposite of the majority.
- Some of the technical trading rules used by contrarians include:
 - Odd-lot theory: Go against volume small trades, opposite to retail investors
 - Cash positions in mutual funds: Go against view of mutual funds
 - Put/Call ratio: Go contrary to put/call volumes
 - Short interest: Go against volume of shares that are short
 - 200-day moving average
 - Advance/decline line: Bet on breadth of market



Investment Theory

- MPT says that portfolio assets should be selected **based on how they interact within the portfolio**, rather than only on an asset-by-asset basis.
- A lot of the assumptions behind MPT may not be valid, but MPT is still a good start point

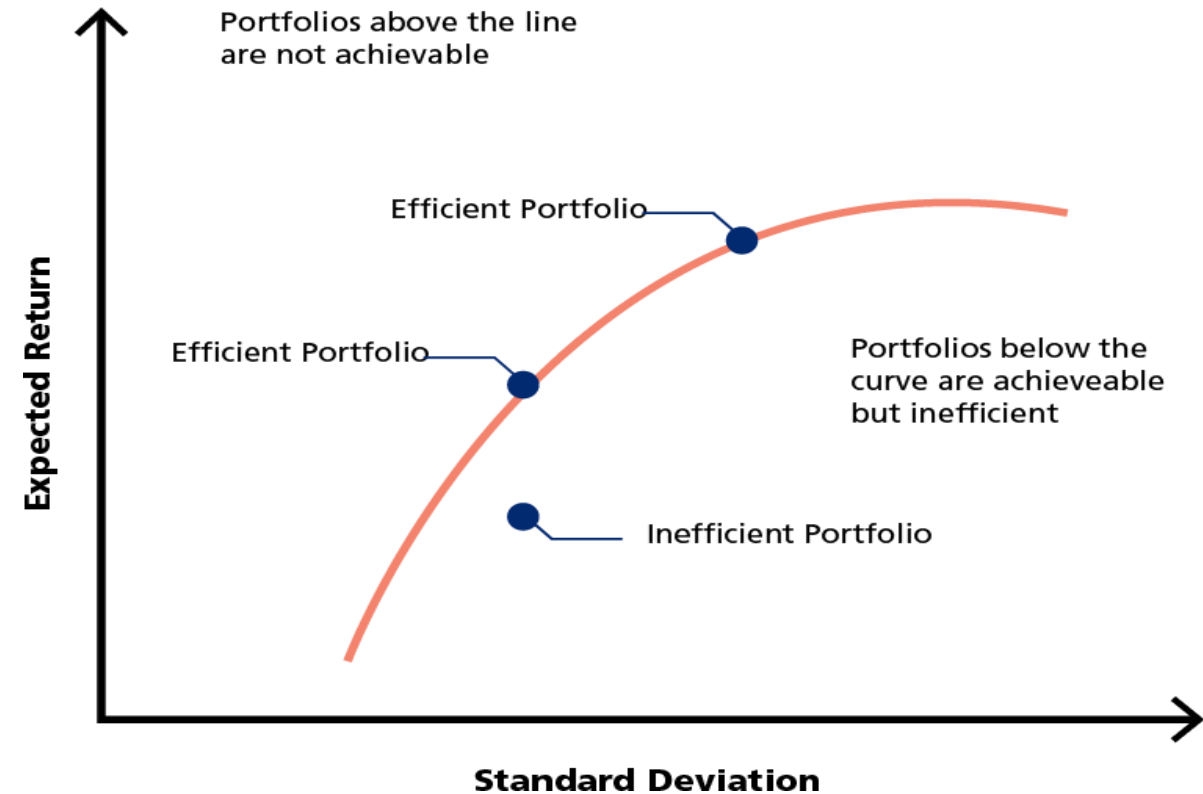


Critique

1. Investors are not always rational
2. Downside risk is also a concern for investors, along with total risk
3. Risk return varies
4. Information is not equal
5. Perfect competition is not true

#1: Efficient Frontier concept (Optimal Portfolios)

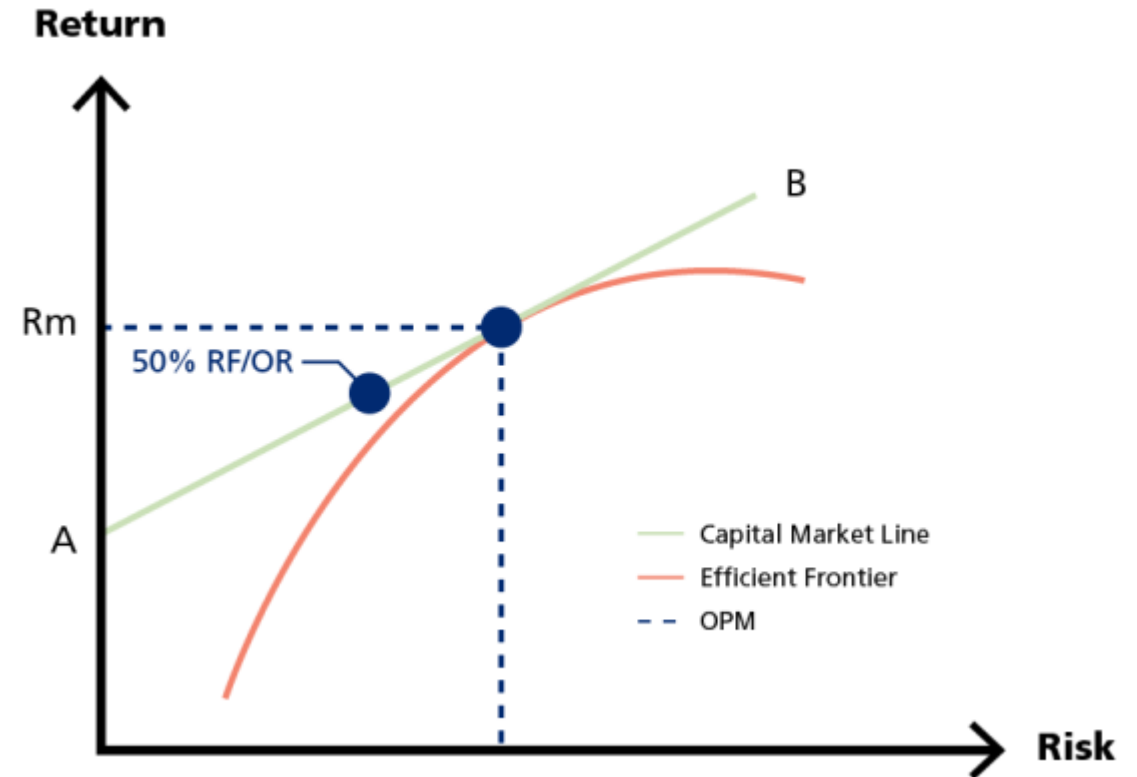
- Builds on the MPT assumption that investors **want highest return for a given level of risk.**
- If we map these optimal risk-return combinations, we would have what MPT calls the **efficient frontier**. E.g.
 - For SD = 1%, 2%, 3% etc we plot the expected return at that SD level
- The investor cannot build an attainable portfolio above the line, and any portfolio below the line is not as efficient.
- **All points on the EF line are equally efficient, and may reflect preferences of different clients**



2: Capital Asset Pricing Model (CAPM) concept

- This concept was developed by William Sharpe, based on the PMT.
- It looks to estimate a required rate of return for any risky asset.
- CAPM, pronounced as CAP-M, has two divisions,
 - **CML**: a broader version focusing on the overall capital market, called the capital market line
 - **SML**: a narrower version focusing on individual securities, called the security market line

- We assume that a risk-free security exists
 - Point A in the graph, i.e. 5% return with SD = 0
- A line (AB) is drawn from this point to the meet the Efficient Frontier curve (red)
- This point where the lines meet corresponds to a return on **R_m**
- Between A and R_m we can create a portfolio with a mix of safe + risky assets
 - E.g. a client chooses 50% Risk Free (RF) at an Optimum Risk (OR)



To calculate the capital market line, use the following formula:

$$r_p = r_f + \sigma_p \left(\frac{r_m - r_f}{\sigma_m} \right)$$

Where:

r_p = portfolio return

r_m = market return

r_f = risk-free rate

σ_p = standard deviation of the portfolio

σ_m = standard deviation of the market

Example:

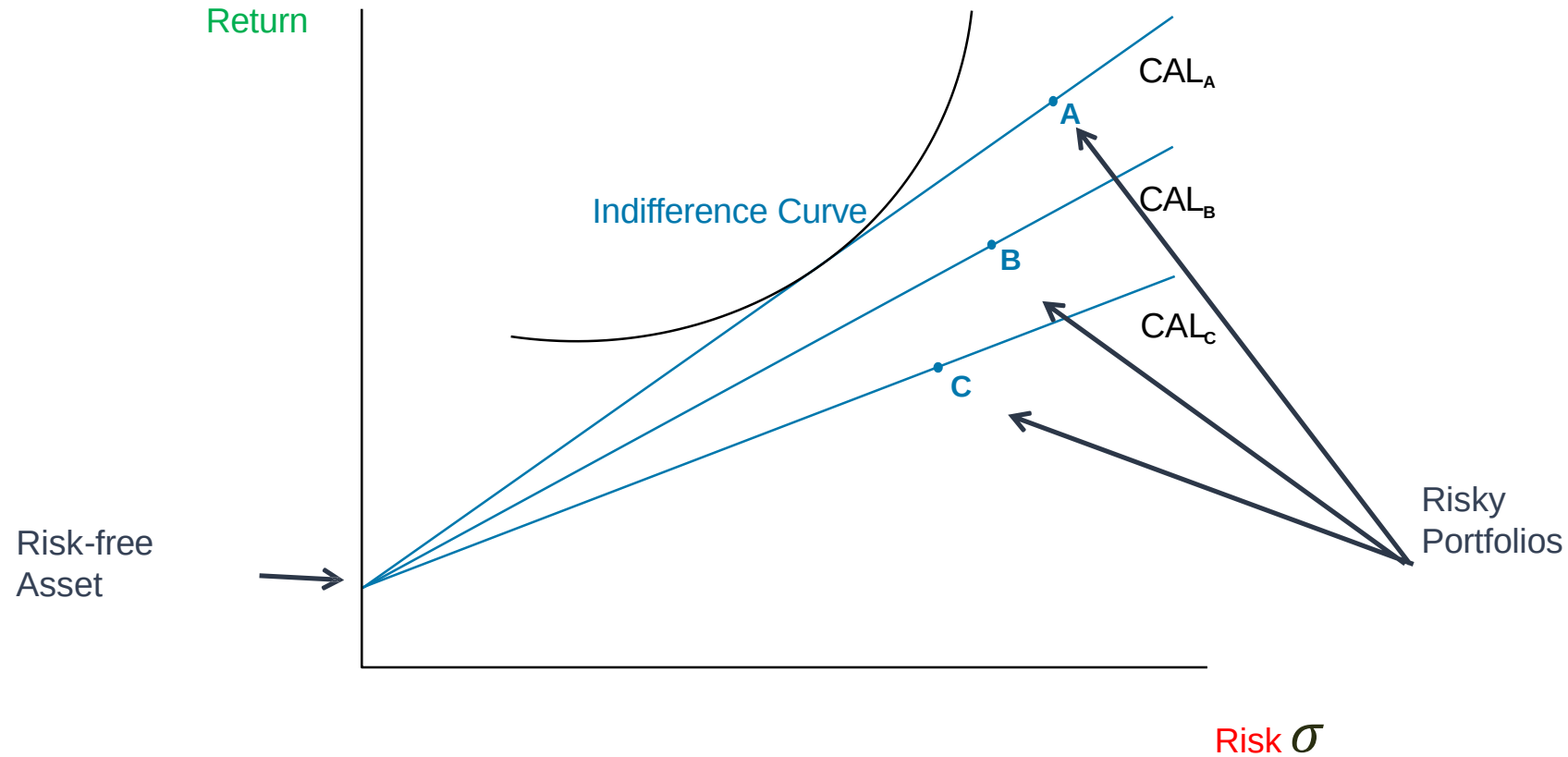
Calculate portfolio return, given

- Market return = 10%
- Risk-free rate = 6%
- SD of portfolio is 3%
- SD of the market is 6%

Solution:

$$\begin{aligned} &= 6\% + 3\% * (10\% - 6\% / 6\%) \\ &= 6\% + 3\% * .67\% \\ &= 8\% \end{aligned}$$

Capital Asset Pricing Model (CAPM)



The Capital Allocation Line. It is a straight line connecting a risk free asset to **ANY** portfolio.

- To calculate the **investor's required return**, start with the risk-free rate. Add to this the relevant risk premium to arrive at the return required for the investor to accept a given amount of risk. The formula for doing this is:

$$R = R_f + \beta_c (R_m - R_f)$$

Where:

r = required rate of return

r_f = risk-free rate

r_m = market rate

- CAPM provides a linear expected return–beta relationship that calculates the expected return given the beta of an asset.
- CAPM makes the transition from total risk to systematic risk, the primary determinant of expected return.

Security Market Line

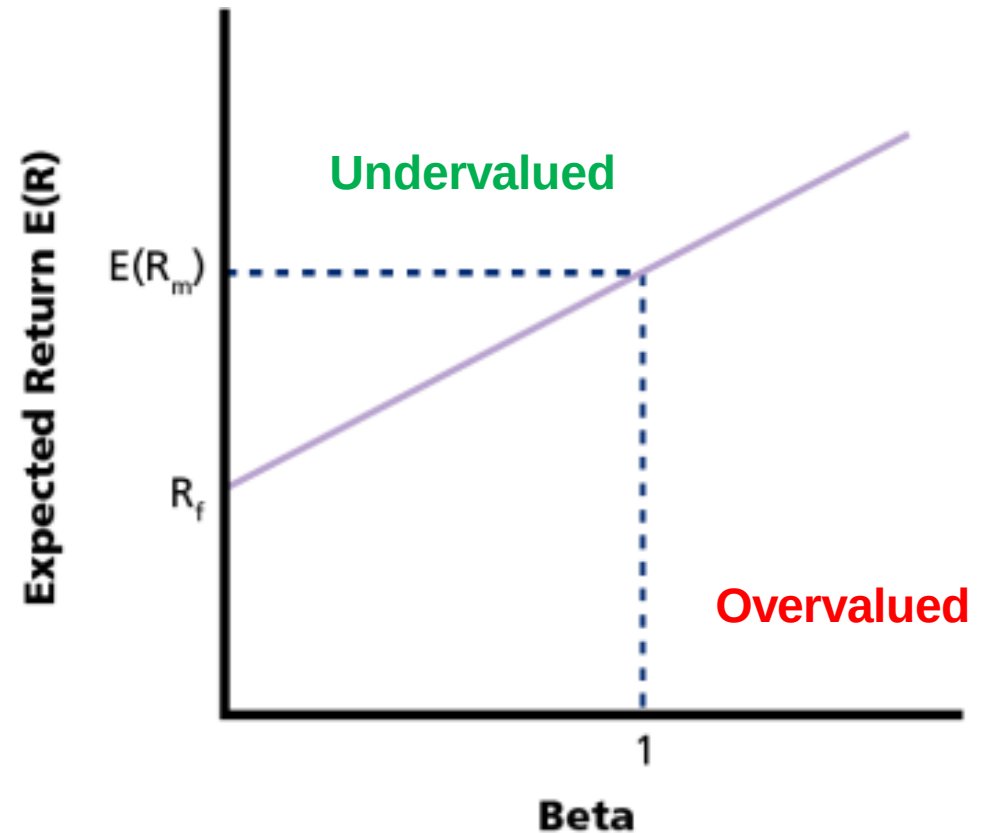
- SML extends CML to securities
- It uses Beta rather than SD

Where:

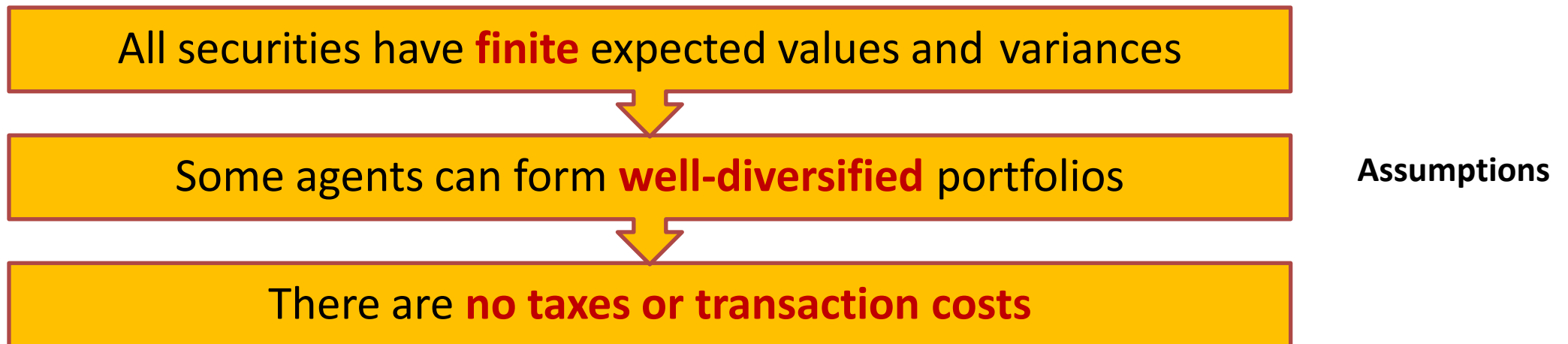
E = expected rate of return

$E(R_m)$ = expected market rate of return

R_f = risk free rate of return



- Is **an alternative model to MPT / CAPM**
- It assumes that various market, macroeconomic and asset-specific factors affect an asset's return. These factors are used to estimate the appropriate price of an asset.
- Generally, there should be no arbitrage, Investors would take advantage of any opportunities created by mispriced securities, thereby creating equilibrium.
- APT is a **multifactor model**, unlike CAPM, which is a single-factor model (beta).



- Fama, in his Ph.D. dissertation called EMH, stated that at any given time and in a liquid market, **security prices fully reflect all available information**
- EMH could be divided into three forms:

Weak

- Assumes current price reflects all information
- Then past price and volume data of a stock have no impact on the future direction of the stock's price.
- **Then technical analysis is not beneficial**, since it adds no predictive value.

Semi strong

- Assumes new information is immediately factored into stock prices by investors and the market regains equilibrium
- the semi-strong form **effectively discounts fundamental analysis.**

Strong

- Assumes current stock prices fully reflect all information, including insider information (i.e., a *perfect* market), and is available to all investors
- Therefore, **one cannot use such information to consistently achieve excess returns.**

- Certain implications arise from acceptance of efficient market hypothesis.
- One of those is that the past movement or direction of the price of a stock cannot be used to predict its future movement
- That future stock prices evolve according to a random walk and thus cannot be predicted with accuracy.
- This is the focus of the random walk hypothesis.
- If random walk is completely valid, one has no reason to do either fundamental or technical analysis.



Asset Allocation

- **Step 1:** Refer the investment policy statement, goals, risk profile, time horizons and other criteria of the client.
- **Step 2:** make asset allocation decisions and choose whether to focus on **active** or **passive** management, or **some of both**.
- **Step 3:** Execute

- The process involves two main decisions:
 1. Which asset classes will be included or excluded from an investor's portfolio.
 2. How the portfolio will be weighted among the asset classes chosen for the portfolio.

Strategic	Tactical	Core/Satellite	Dynamic
• Long-term • Passive once the asset mix has been chosen • Rebalanced periodically to return to the original allocation percentages	• Shorter-term • Responds to changes in the market(s) • Can involve market timing and sector rotation	• Blend of Strategic and Tactical (usually 80%:20%) • Core often invested passively in index funds (beta) • Satellite actively invested (alpha)	• Actively changed in response to market • Primarily a hedging allocation to insure against too large a decline in value • Used mostly by institutional investors

1. Strategic Allocation

- focuses on selecting a mix of different asset classes, assigning a target weighting, guided by modern portfolio theory and creation of optimal portfolio for each client
- rebalancing on a periodic basis (at least annually) to bring portfolio back to target.
- ignores short-term market volatility, hence not a buy-and-hold strategy.

Core Portfolio	SECURE		INCOME		CONSERVATIVE		BALANCED		GROWTH		AGGRESSIVE GROWTH	
Asset Allocation	Range	Target Mix	Range	Target Mix	Range	Target Mix	Range	Target Mix	Range	Target Mix	Range	Target Mix
Cash	0%-35%	20%	0%-35%	5%	0%-25%	5%	0%-20%	5%	0%-20%	5%	0%-20%	5%
Fixed Income	65%-100%	80%	65%-95%	80%	45%-75%	60%	25%-55%	35%	5%-35%	20%	0%	0%
Equities	0%	0%	5%-25%	15%	15%-55%	35%	35%-75%	60%	55%-95%	75%	80%-100%	95%
Total		100%		100%		100%		100%		100%		100%

2. Tactical Allocation

- differs from strategic asset allocation as it attempts to predict which asset classes will be the next to have gains/outperform other asset classes in the short term.
- Funds are then allocated to those asset classes in anticipation of future gains.
- This strategy boils down to security selection and market timing.

	Investment Strategy	
	Strategic Asset Allocation	Tactical Asset Allocation
Goal:	Definition of the optimal long-term investment strategy (5-10 yrs)	Taking advantage of short-to medium term opportunities (0-3 yrs)
Assumptions:	Markets are efficient, assets are at fair value	Markets are inefficient, (some) assets are not at fair value
Application:	Definition of benchmark	Active management
Required Input:	Long term risk- and return assumptions	Models to identify mis-valuation of assets

3. Core/Satellite Allocation

- achieved by dividing a portfolio into a **core** and a **satellite** holding
 1. Core typically represents 70–80% of the total portfolio (i.e., traditional portfolio)
 2. Satellite is 20-30% of the portfolio and is allocated to a tactical strategy (active management) to capture some excess gains (alpha) from the outer ring (yellow assets)



4. Dynamic Allocation

- Dynamic asset allocation is **used primarily by institutional investors.**
- This type of allocation is used primarily **to hedge a portfolio** against big declines in value.
- To do this, the investment manager will shift between riskless and risky assets, making the changes in response to what is happening in the marketplace.
- As portfolio values change, the allocation will change with it.
- This **requires more-or-less constant oversight**, which is something institutional investors have the ability to do.

- Portfolio will require periodic rebalancing to maintain its original focus and intention.
- The goal will be to bring the portfolio back to its desired allocation, by selling some assets that have strong growth and adding others that have not done as well.
- Rebalancing **is primarily about risk control**, or making sure the portfolio isn't overly dependent on the success or failure of one investment, asset class, or style.

Time-based

- Fix a frequency, e.g. quarterly, half yearly or at least annually
- **Act if needed**

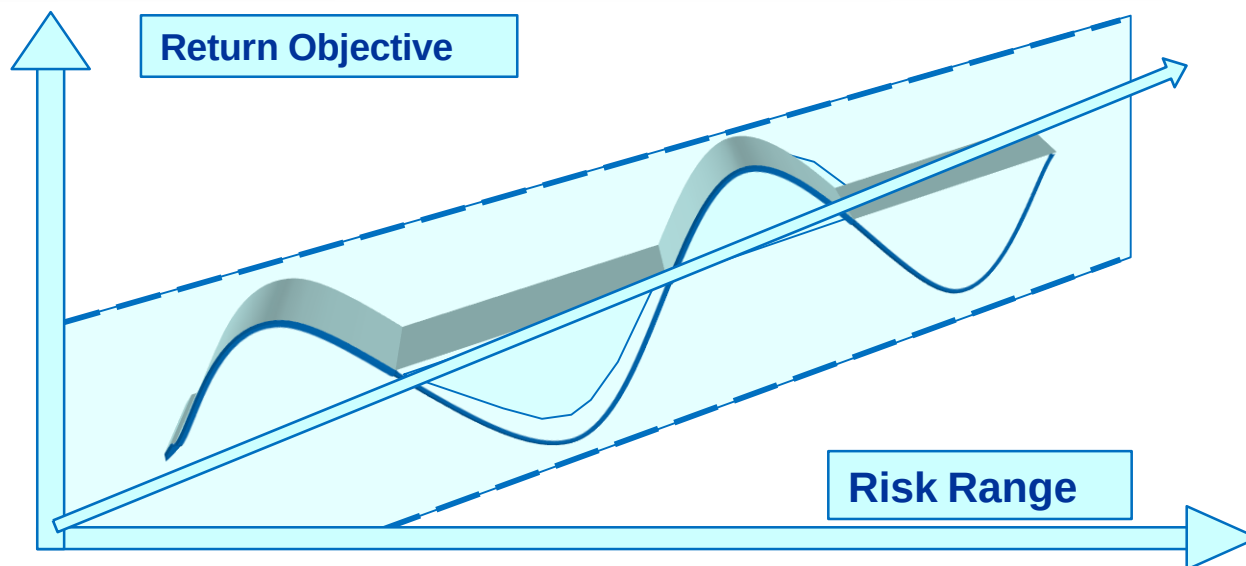
Threshold-based

- Define **upper or lower thresholds for each asset**, e.g. $\pm 15\%$
- **Act only if threshold is met**

Both

- Fixed date, typically annually,
- Then rebalance based on thresholds

Examples: Rebalancing



When rebalancing occurs, gains from over-weighted positions are sold (locking in some of the profit) and used to buy shares of the underweighted positions (at a cheaper price), which effectively forces a buy low, sell high approach.

Example – Selling & buying to rebalance



Example – SIP with no rebalancing



- We looked at Jensen's alpha, and how to determine the degree to which an asset manager is adding value to the portfolio.
- The consideration of alpha brings us to the next portfolio decision: active or passive management.
- **Active management**, as the name implies, involves **actively selecting securities** based on the belief that doing so will allow the asset manager to achieve higher returns (after fees) than those produced by the market.
- **Passive is conversely, based on less is more**



- Active management comes with both advantages and disadvantages
- However, **for most it is a hit or miss in terms of consistency**, i.e.
 - When it works, it can work quite well
 - When it works, the manager feels pretty smart and tries to replicate again
- **Some active managers can consistently succeed**. E.g. Warren Buffett, Peter Lynch, Sir John Templeton. **Peter Lynch's suggestions are:**
 1. Keep return expectations realistic (neither too high nor too low).
 2. If the best you can do is match market performance, buy index funds.
 3. Long-term investing is generally more effective and efficient than short-term trading
 4. Adapt the portfolio to changing client requirements, as necessary.
- Passive management generates lower fees than active management

1- LADDER

- A bond ladder is a portfolio of individual bonds with staggered maturities, for eg: The first bond could mature in one year, the second in two years, the third in three **years, and so on.**
- As each bond matures, the money is used to purchase a new bond with a five-year maturity.
- Using this process, interest rates would be locked in for each five-year period, but the entire amount is never subject to long-term interest rate (reinvestment) risk.
- Laddering also helps **maintain liquidity**, because on maturity amount becomes available each year (rather than the entire amount being locked up for the full period until maturity).

2- BULLET

- A bullet portfolio holds almost all its bonds in **intermediate maturities**, rather than spreading the maturities out.
- Investors may hold a small amount of short-term (1-3 year) bonds and long-term bonds (20 years), but the majority will be intermediate term (10 years).
- An investor or advisor opting for a bullet portfolio is effectively “betting” that during the target maturity window interest rates on offer for reinvestment will be relatively attractive.

3- IMMUNIZATION

- Bond immunization is **an investment strategy used to minimize the interest rate risk of bond investments by adjusting the portfolio duration to match** the investor's investment time horizon.
- The immunization process helps to protect the portfolio against decreases to principal because of interest rate increases.
- **For example:** If the client wants to retire in 10 years, bond portfolio duration could be adjusted to match the 10-year time frame. This would help immunize the portfolio against interest rate and reinvestment rate risk.

4- BARBELL

- Barbells purchase **short-term and long- term bonds** (but no mid-term bonds).
- The idea is to lock in higher interest rates with longer-term bonds, while the short-maturity bonds provide flexibility and liquidity.
- If rates fall, when they mature, the short-term bonds can be invested in other assets with higher returns.



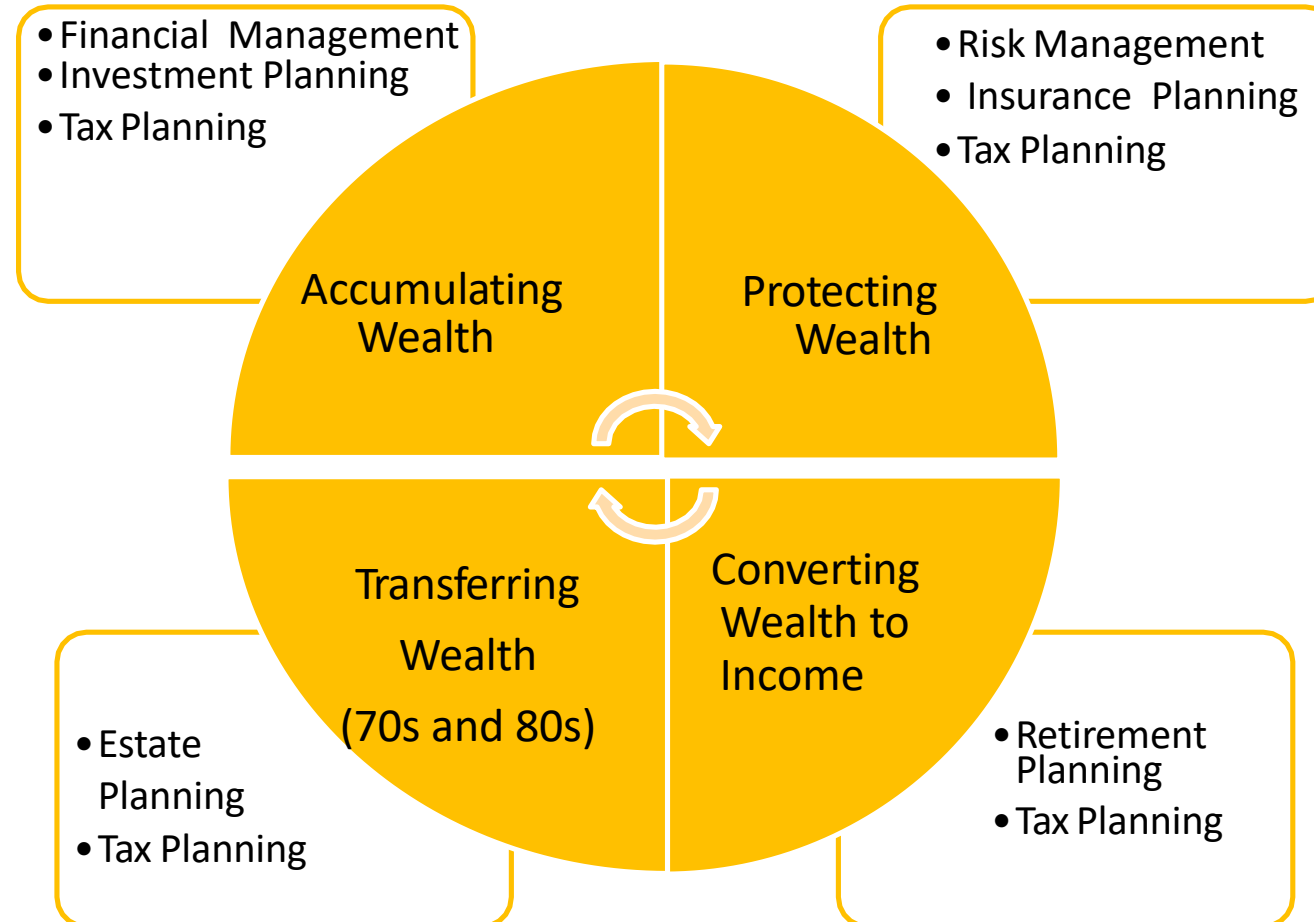
Wealth Management

1. Financial Management
2. **Investment Planning**
3. Insurance and Risk Management
4. Retirement Planning
5. Tax Planning
6. Estate Planning

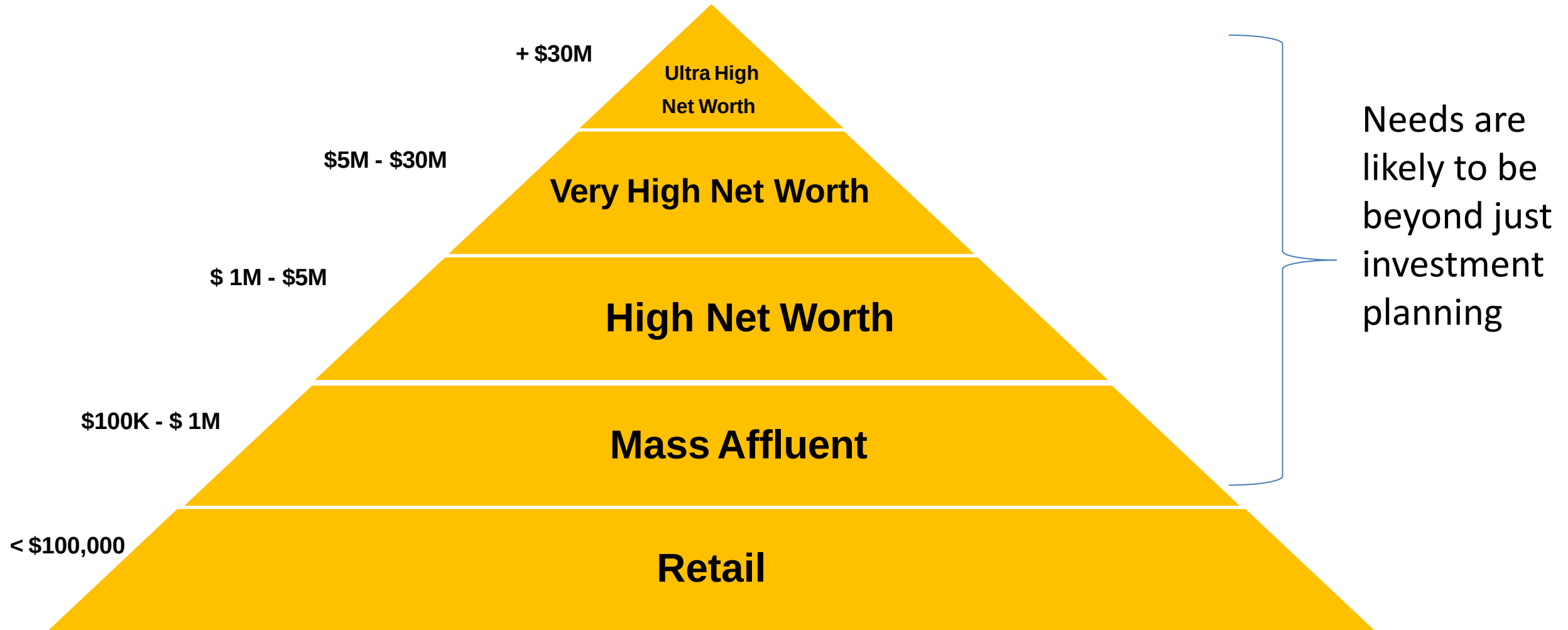
Wealth Management covers both the ongoing investment planning / asset management as well as more holistic goal assurance



- Four financial phases of wealth management are as follows



Levels of Wealth (USD example)



- It's important to recognize that HNWIs have many of the same needs and concerns as those with lesser means.
- It's just as important to remember that those needs and concerns often expand beyond the basics into more specialized territory.
- Such HNWIs may be offered these services by a **personal Wealth manager** or **via their bank** (i.e. private banking services)
- To effectively serve HNWIs, **investment advisors must educate themselves** on the **special needs and requirements** of this market segment.
- **Further, positioning oneself as a wealth manager demands a high level of investment-related acumen, along with significant real-world experience.**

Investments and Property	+ Own company	+ Buying a company	+ Stock options and similar	+ Alternative investments
• Retail assets • Own house(s)	• Succession planning (process) including valuation	• Valuation	• Strategies relating to hold / sell	• Managed Futures • Trusts • Foundations

- For many individuals, specific planning is required for
 1. Succession planning for own business
 2. Concentrated ownership in companies
 - Via investing in a private company
 - Via stockholding in a public company

Tasks for managing succession / concentrated holdings

Family Succession	Strategic Buyers	Management and Financial Investors
1. Conduct an annual assessment of the near-term and longer-term viability of the family business.	1. Conduct research to identify a sufficient number of qualified buyers.	1. Understand the sources of financing available and what different types of financial investors look for in a company.
2. Ensure that key points of shareholder agreements are clearly documented and signed off by all parties.	2. Control the amount and nature of information conveyed to buyers.	2. Recognize the payoffs and risks of using debt to finance a management buyout (MBO).
3. Implement a governance structure that promotes transparency and accountability	3. Consider the terms of the deal as well as the stated purchase price.	3. Actively manage the conflict of interest that MBO's create for management.
4. Manage the transition with management, customers and suppliers.	4. Ensure that the letter of intent (LOI) is clear and comprehensive.	4. Consider the ability of a financial investor to be a good business partner as well as the amount they are willing to invest.
5. Create a mechanism to afford liquidity for family members desiring an exit.	5. Manage the operations of the business through to closing.	5. Understand the financial dynamics of an MBO and how all parties can realize value in the process.

- Financial statement ratios are useful as a quick way to compare an aspect of one business with others like it.
- **Ratio analysis** can be classified under four major headings:
 - **Liquidity ratios**, determine whether a company can **meet its obligations** as they come due.
 - **Activity ratios**, determine how **rapidly assets move** through the company, including how quickly inventory is moving out the door, and how quickly payments are received.
 - **Profitability ratios**, measure performance and **how profitable** the company is, and what sort of return the company is getting on capital and equity.
 - **Leverage ratios**, such as debt-to-equity and debt-to-capital, measure the extent to which a company **uses debt financing**.

Liquidity ratios measure the ability of a company to meet its current financial obligations, i.e.

- how many times are a company's current liabilities covered by current assets and
- components of current assets

- **Current ratio**
$$\frac{\text{Current assets}}{\text{Current liabilities}}$$

- **Quick ratio (acid test)**
$$\frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}}$$

- These are also known as operating efficiency ratios and measure how well the company is utilizing its assets.

- **Inventory turnover**
$$\frac{\text{Sales}}{\text{Average inventory}}$$

- **Accounts receivable turnover**
$$\frac{\text{Credit sales}}{\text{Accounts receivable}}$$

1. **gross profit** (gross profit margin)
2. **operating profit** (operating profit margin)
3. **net profit** (net profit margin).
4. **EBIT** (earnings before interest and taxes) can be considered equivalent to operating profit margin.
5. **EBITDA** (earnings before interest, taxes, depreciation and amortization) can be considered equivalent to cash flow from operations.

Net Sales	
Less	Cost of sales
Equals	Gross profit margin
Less	Operating expenses
Equals	Operating profit margin
Less	Income taxes and interest expense
Equals	Net profit margin

- The ROE is one of the most important ratios because it reveals a lot about how well a company is managed.

$$\text{ROE} = \frac{\text{Earnings after taxes} - \text{Preferred dividends}}{\text{Equity} - \text{Preferred stock}}$$

- Leverage ratios show the percentage of total capitalization (debt plus equity) that is debt.
- Several ratios are used, but the ones that analysts watch most closely are the **debt-to-total-capital** and the **debt-to-equity ratios**.
- Long-term debt is used in the numerator because it is not paid off during a normal operating cycle.
 - Therefore, it can be manageable today, assuming a favorable business environment, but it could become a burden when the business cycle changes direction.
 - The most straightforward description for the figure in the denominator is total capital, long-term debt plus shareholders' equity.

- **Debt-to-total-capital ratio**

$$\frac{\text{Long term debt}}{\text{Total capital}}$$

2. Discounted Cash Flows

- The DCF methodology consists of 6 steps:
 1. forecast the **discretionary cash flow** that a company is expected to generate in each of the next few years, generally three to five years.
 2. **discount to a present value** amount using a discount rate
 3. calculate **terminal value** by estimating the annual discretionary cash flow beyond the forecast period, divided by a capitalization rate.
 4. **Discount the terminal value** to a present value amount using the same discount rate applied to the discretionary cash flows
 5. **Add** the present value of the terminal values to the present value of the discretionary cash flows to calculate the **enterprise value of the company**
 6. Interest bearing debt is deducted from enterprise value to **derive the shareholder value of the company.**

3. Executive Stock Options

- Executives can have a high concentration of their portfolio invested in one company.
- A stock option gives the holder the right to purchase a share of company stock at a particular price (i.e., strike price) for a set period.
- According to a paper by Morgan Stanley, the first step to address a concentrated stock position is to define strategic objectives, which may include:
 - Creating liquidity
 - Hedging risk
 - Diversifying the concentrated position
 - Accepting the risk of a concentrated position

Create Liquidity	Hedge	Diversify
Variable prepaid forward contract	Variable prepaid forward contract	Variable prepaid forward contract
Sell the position	Protective put and put spread options	Sell the position
Covered call	Zero-premium collar	Exchange (sell over time)
Borrow against the position	Stop loss orders	

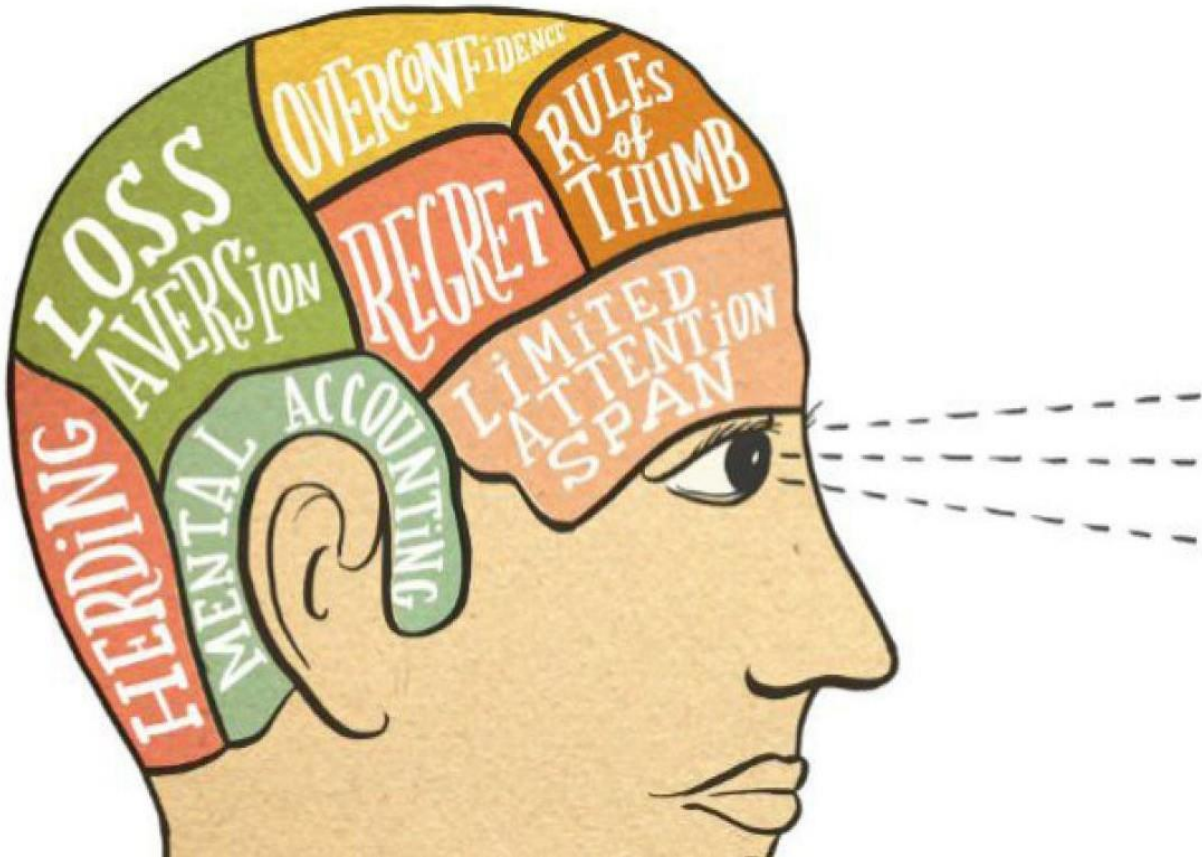
4. Alternative Investments for High Net Worth Individuals

- The use of alternatives opens the portfolio to investment possibilities in asset classes outside of those normally considered to be traditional.
- These may be for a variety of goals, e.g,
 - Tax purposes, Better returns, Estate planning, Charitable objectives etc
- Vehicles include
 - Managed Futures
 - liquid alternative investment strategy where professional portfolio managers called Commodity Trader Advisors (CTAs) invest in futures contracts.
 - Trusts
 - For advanced tax planning and more complex estate planning issues
 - Foundations and endowments
 - entities created for philanthropic and charitable activities.

- Private banking is an alternative to nonbank services offered to high net worth individuals (HNWIs).
- Private bankers usually offer investment-related advice and specialized financial solutions.
- Some of the available services include special access to loans and lines of credit, estate and retirement planning, and access to investment alternatives.
- Most private banks require a relatively high level of investable assets (e.g., \$500,000 and more).
- Banks also generally provide a high level of privacy and confidentiality to clients.



Behavioral Finance



Behavioural Finance

- Science of applying psychology to finance.
- Disputes the notion that markets are efficient and rational.
- Suggests that Human Beings are not always rational and do not always act in a rational manner.
- The impact of this on financial markets, and more specifically, on individual investors is significant.

1. Research has shown **that people are affected by internal biases**, resulting from hereditary or experiential reasons.
 - 21 investor biases exist in three categories:
 1. general behavioral biases (6),
 2. information processing biases (8) and
 3. emotional biases (7)
2. Similarly, **money affects a person's thinking and behaviour**
 - 8 additional effects of money
3. Hence, an advisor needs to **apply behavioural finance techniques** to better manage client and their personality

Cognitive Dissonance:

- When newly acquired information conflicts with a pre-existing notion or understanding, people will often perform rationalization to maintain psychological stability.

Conservatism:

- People cling to prior views or forecasts at the expense of acknowledging new information
- Results in specific denial of, or under-reaction to new information

Confirmation:

- Refers to a type of selective perception that emphasizes ideas that confirm our beliefs, while devaluing whatever might contradict them.

Representativeness:

- People often apply prior life experiences to new situations. Sometimes this works, but other times it may lead to making significant financial errors.

Illusion of Control:

- Tendency to believe that one can control or influence outcomes, when this is not possible in many circumstances.

Hindsight:

- People tend to overestimate the accuracy of their own predictions and underestimate actual levels of uncertainty.

Anchoring and Adjustment:

- rather than deciding based on pure value, we factor in comparative value

Mental Accounting:

- Tendency of grouping assets into non-interchangeable mental accounts.

Framing:

- Tendency to respond to various situations differently based on the context.

Availability:

- people estimate the probability of a new outcome based on its familiarity in their lives.

Ambiguity Aversion:

- People tend to dislike uncertainty (or ambiguity) more than they dislike risk.

Self Attribution:

- tendency to ascribe successes to oneself, while blaming failures on outside influences.

Outcome:

- Deciding based on the outcomes of events, without researching about the steps leading to it

Recency:

- causes an individual to place more emphasis on recent events than those that occurred in the past.

Loss Aversion:

- Person places more emphasis on recent events than those that occurred in the past.

Overconfidence:

- An unwarranted faith in one's reasoning, judgement and cognitive abilities.

Optimism:

- The tendency to have unrealistically positive view of themselves and their futures.

(Lack of) Self Control:

- The tendency to consume today rather than save for tomorrow

Status Quo:

- Bias to maintain the existing state of affairs

Endowment:

- demand a higher price to sell an owned asset, and a lower price to purchase an unowned asset.

Regret Aversion:

- The fear that any decisive action may prove to be less than optimal.

Money often generates strong feelings in people

- As we saw with behavioural biases, the way people feel about money often influences them to make money decisions that are not necessarily in their best interest.
- To complicate matters, most people are not aware of their biases or money psychology

Money psychology - Personality types:

1. Hoarders
2. Spenders
3. Bingers
4. Money Monks
5. Avoiders
6. Amassers
7. Worriers
8. Risk Takers

Hoarders:

- Enjoy holding onto their money and struggle to spend it

Spenders:

- Use money to buy whatever is assumed to bring pleasure.

Bingers:

- Are a combination of hoarders and spenders.
- They tend to save, and then spend the money all at once.

Avoiders:

- Tend to avoid performing various everyday money management tasks.

Worriers:

- Tend to worry excessively about money.
- They typically want to exercise a great amount of control over their money

Risk Takers:

- Enjoy taking money risks.

Helping Clients to “Behave” Their Way to Success

- The way to help clients behave their way to success is to help them **recognize and revise** at least some of their **core disabling biases**
- However, it is quite difficult to get people change even one of their behavioural biases.

What can you do to help?

1. Try to follow a disciplined investing approach.
2. Help the investor focus on goals and achievements
3. Create a disciplined approach to investing, rebalancing and evaluating progress
4. Keep client biases from negatively impacting the goals–achievement.
5. Some advisors also suggest the client take a small percentage (e.g., 10%) of their money and make their own investments



Personalities and **Risk profiles**

Ethics in Financial Planning: 8 Principles

Principle 1: Client First

- Place the client's interests first.

Principle 2: Integrity

- Provide professional services with integrity.

Principle 3: Objectivity

- Provide professional services objectively.

Principle 4: Fairness

- Be fair and reasonable
- Disclose and manage conflicts of interest.

Principle 5: Professionalism

- Demonstrate exemplary professional conduct.

Principle 6: Competence

- Maintain the abilities, skills and knowledge necessary to provide professional services competently.

Principle 7: Confidentiality

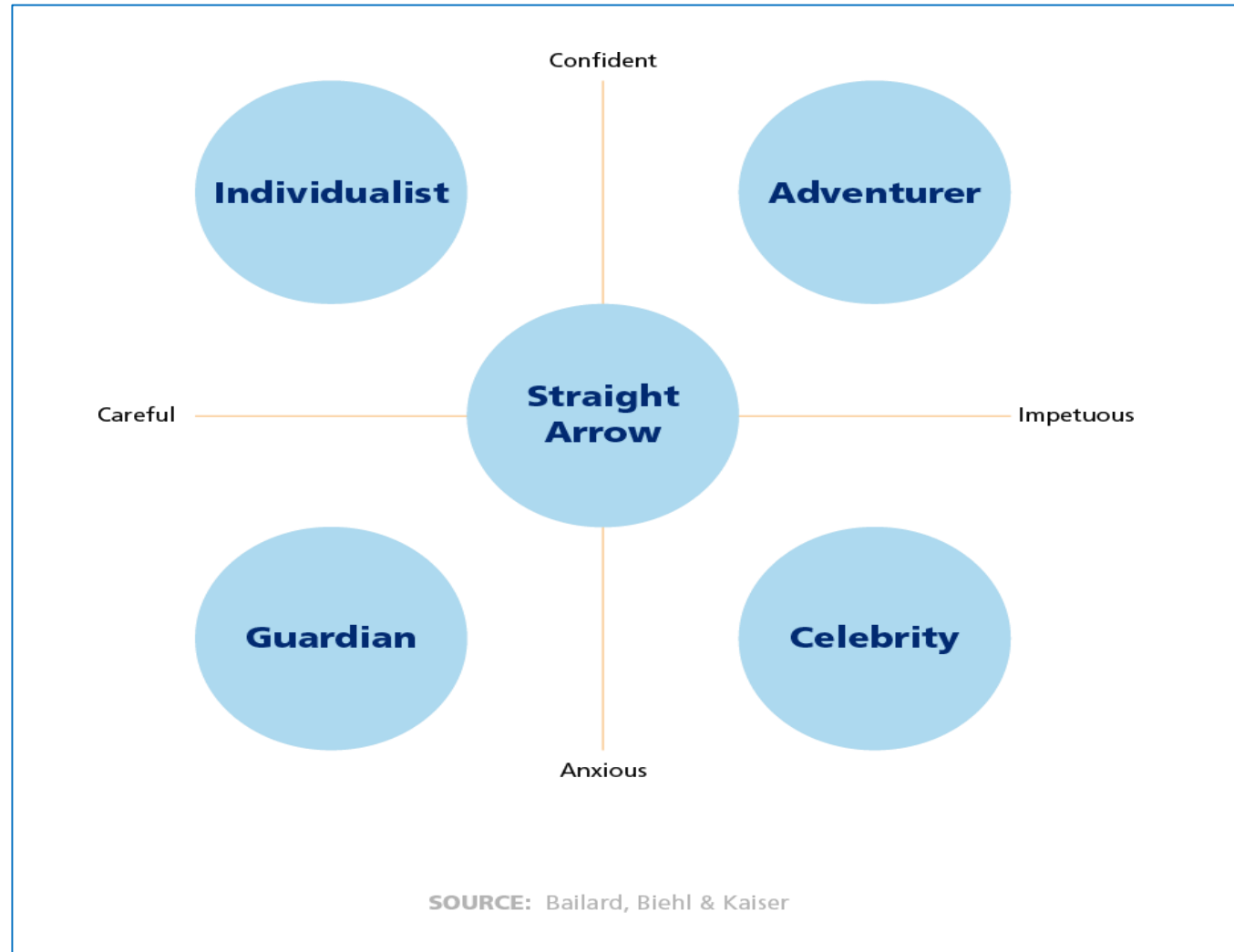
- Protect the confidentiality of all client information.

Principle 8: Diligence

- Provide professional services diligently.

Investor Personalities

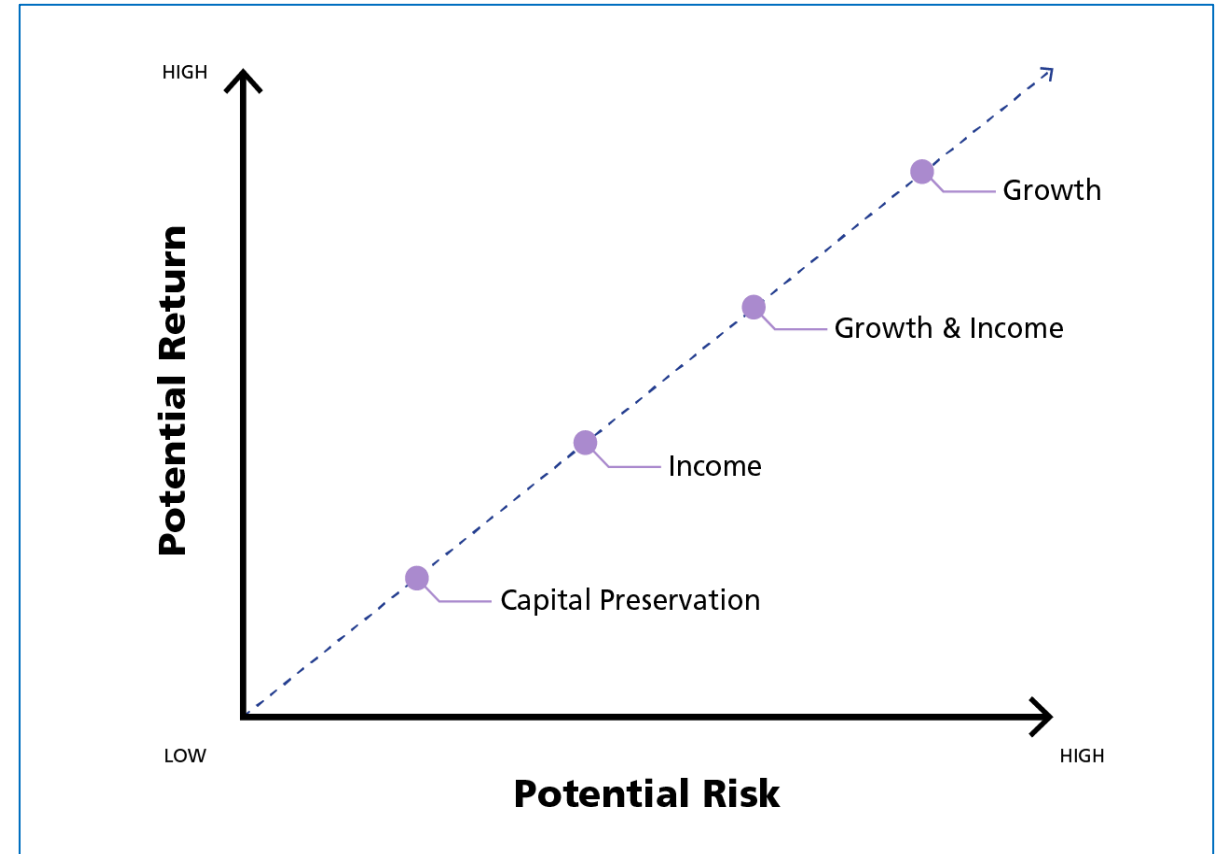
- Client's display certain personality traits that give insight into the type of investor that they are or are likely to be.
- Investor personalities are an important factor in the creation and maintenance of a happy working relationship.
- Personality can be mapped on axis of **careful-impetuous** and **anxious-confidence**



Risk Tolerance and Suitability

- All financial planning involves a **risk-return trade-off**
- To increase return from investment, the risk level needs to be increased.
- Taking higher risk is not a guarantee of higher return.
- Hence it is essential that
 1. the clients **risk profile** (i.e. tolerance for risk) be understood (usually via questionnaire)
 2. A **sustainable plan** for the same is built

End-result: Client is mapped to **one core objective**



Risk Tolerance Questionnaires

- Many risk tolerance questionnaires exist and are used by Financial Professionals.
- Questions include
 - How much of a loss in a year would you be willing to endure without liquidating your portfolio?
 - How soon would you need to access your funds?
 - How stable is your financial situation?
 - How often do you evaluate your portfolio's performance?
 - Of the following, which type of stock (or bond) would you choose (with a list of categories from which to choose)?
 - What is your net worth (and income)?

Suitability: Using Risk Tolerance Information

- Risk tolerance normally falls into 3 categories:
conservative / moderately aggressive / aggressive
- These typically translate **into four primary investment objectives**/strategies and **two sub-strategies**/categories.
 1. Capital Preservation
 2. Income
 3. Income with Moderate Growth
 4. Growth with Moderate Income
 5. Growth
 6. Aggressive Growth

Understanding Life Stages

- A client's goals make more sense when looked at in terms of life stage
- Each stage contains different priorities and focus, such as:

1. Accumulation Phase (to about age 35)

- First job/career, Paying off debt, Begin saving and investing

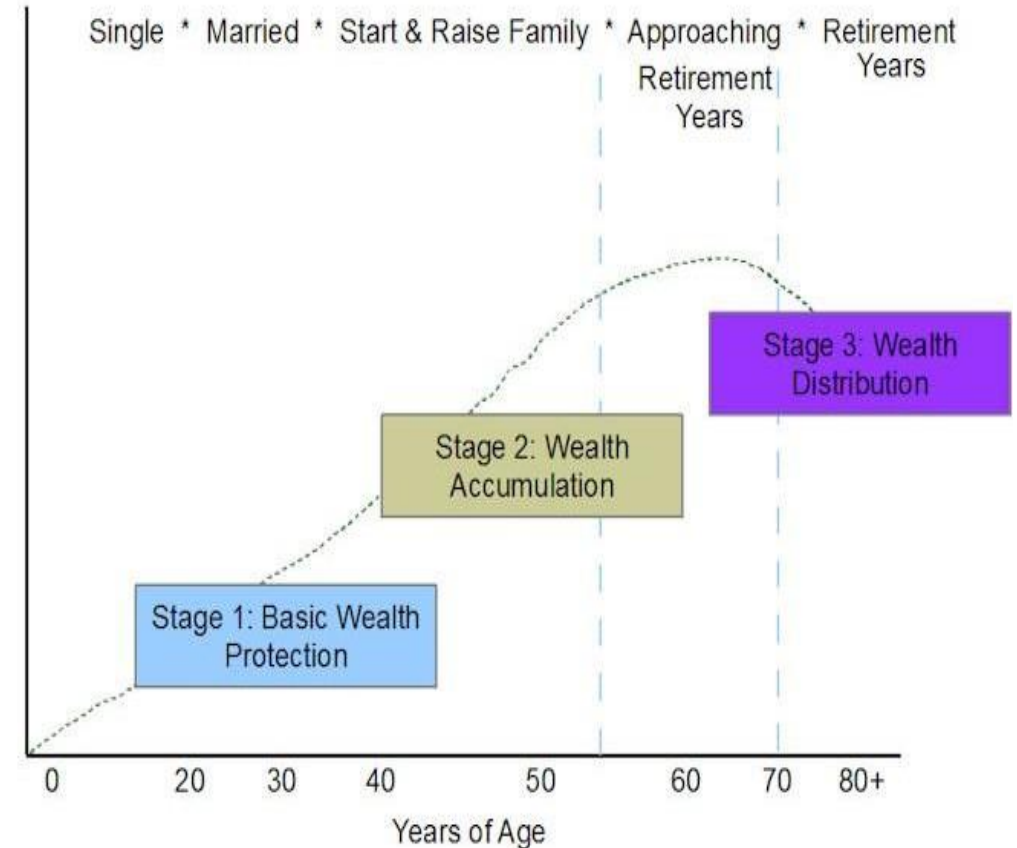
2. Consolidation Phase (generally age 35 to 60)

- Home purchase, Marriage, Children's educations, Peak earning years, more investible income

3. Retirement phase (generally age 60+)

- End of work income, shift to more conservative expenses, Portfolio distributions, Wealth Transfer, Long-term care issues

An Individual's Financial Life Cycle



Investor emotions and life stages

Barclays: Average Investor Emotions Can Affect Investment Returns



Investment Policy Statement

- Lays out the roadmap for a client's situation, their goals/objectives, and the investment strategy or approach, used to achieve those goals. a good IPS will contain most of the following items:
 1. Executive Summary that provides an overview
 2. Client Description
 3. Investment objectives and client goals
 4. Fiduciary responsibilities
 5. Client responsibilities
 6. Investment constraints
 7. Investment philosophy
 8. Asset allocation strategy
 9. Security selection criteria
 10. Implementation, monitoring, and follow up
 11. Review Schedule